

Heterogeneous Collaborative Pursuit via Coverage Control Driven by Fokker-Planck Equations

Ruoyu Lin¹, Soobum Kim², and Magnus Egerstedt¹

Abstract—Inspired by common features found in collaborative behaviors in nature, we investigate a general collaborative pursuit framework enabling heterogeneous multi-robot systems to adapt to dynamic environments and diverse tasks. A class of augmented Fokker-Planck equations is formulated to characterize dynamic environmental conditions, and the resulting time-varying density functions drive a novel coverage-based controller, with provable stability properties, for the participating robots to perform tasks in real time. The developed framework is decentralized and incorporates heterogeneity among different robots in task suitability, relative performance in a specific task, and safe operating regions. To demonstrate its adaptivity and effectiveness, the framework is implemented across four experimental applications ranging from multi-robot coordination to collaboration, namely forest firefighting, pursuit-evasion, monitoring of various environmental phenomena, and phoretic interactions.

Index Terms—Multi-robot systems, cooperating robots, distributed robot systems, networked robots.

I. INTRODUCTION

COLLABORATIVE strategies are essential to survival in nature, as they allow a group of individuals to accomplish tasks that are difficult for a single agent to achieve. For example, a group of reindeer forms a swirling circle when threatened in order to protect themselves from natural enemies, e.g., [1], a collective of ants can carry objects that a single individual cannot, e.g., [2], and a team of geese flocks in a V-formation to extend their migratory reach, e.g., [3]. These group-level collaborative behaviors typically emerge from individual-level rules governed by local principles [4], [5].

Drawing inspiration from such biological behaviors, swarm robotics techniques have been developed in various applications such as wildfire monitoring, e.g., [6], pursuit-evasion, e.g., [7], search and rescue, e.g., [8], and collaborative transport, e.g., [9]. However, despite the inherent robustness of

multi-agent systems to individual failures, [10], [11], adaptivity has not been quite as straightforward to achieve. This is the main topic under consideration in this paper and, in particular, we consider adaptivity from two vantage points:

- (A) The agents can respond dynamically to changing environmental conditions in real time.
- (B) The group-level behavior, based on local interaction rules, can accommodate diverse tasks without having to redefine the structure of the interaction rules themselves.

To achieve the first type of adaptivity (A), we draw on the idea proposed in [12] where an ecologically inspired agent tends to minimize its effort subject to liveness [13] and safety constraints. The framework in [12], [13] can be inherently adaptive to changing environmental conditions as control decisions for the participating agents are made at each moment in time, which is different from more standard optimal control approaches that require the control decisions to be solved before the agents take actions and often result in fragility in the sense that dynamically changing environmental conditions can easily invalidate a priori models [14].

To achieve the second type of adaptivity (B), one first needs to examine common drivers of collaborative behaviors in nature. One such driver is a set of ecological pressures or constraints, imposed by the environment on the organisms [15]. This driver consists of two main components: the interactions between an agent and its neighbors, and the interactions between the agent and its local environmental conditions, such as fluid dynamics and temperature changes for fish, aerodynamics and light conditions for geese, and pheromones and terrain conditions for ants. Several of these environmental conditions or physical phenomena can be described by certain types of partial differential equations (PDEs), e.g., [16]. Thus, we define the environmental conditions as two hierarchical concepts, namely the physical quantities governed by certain PDEs that interact directly with the agents dynamically, and the environmental factors that only directly affect these physical quantities. If we think of the PDEs as describing the environment, the agents can be related to the solutions to them through a coverage control framework, e.g., [17]. As such, each agent's behavior is driven by the local environmental conditions, characterized by certain PDEs, and the behaviors of its neighbors, encoded in a coverage control framework.

Moreover, since there exist interactions between agents and their environmental conditions, not only can the environmental conditions drive the behavior of agents, but the agents can also influence their environmental conditions. Hence, we will propose a general agent-environment interaction framework

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through the context of predator-prey interactions in this paper, where the terminology “prey” can capture different concepts such as energy, information, environmental phenomena, and user-specified principles to guide the behavior of agents.

The main contributions of this paper are as follows.

- (1) We propose an ecologically inspired multi-robot collaborative pursuit framework that is adaptive to not only dynamic environments but also diverse task scenarios. The proposed framework is decentralized based on a novel coverage control strategy that governs the behavior of the robots and is driven by a class of PDEs encoding dynamic environments. In total, eleven heterogeneous modalities related to the agent-environment interactions are encapsulated within this framework.
- (2) The novel coverage control strategy we develop, with proved stability, enables heterogeneous robots to optimally cover multiple distinct time-varying density distributions, where the heterogeneity in task suitability is encoded by the predator-prey graph defined in Section III, and the heterogeneity in relative performance among different robots in a specific task is encoded by the multiplicatively weighted Voronoi diagram.
- (3) Based on the fundamental conservation law of nature, a class of PDEs, which encode dynamic environmental conditions and are augmented by the dynamics of participating robots, is constructed building upon the form of Fokker–Planck equation, with its asymptotic stability proved under certain conditions.
- (4) The effectiveness of the proposed adaptive collaborative pursuit framework is demonstrated through four applications in experiments, namely forest firefighting with a team of heterogeneous robots, pursuit–evasion with heterogeneous pursuers and evaders, multiple robots with complementary sensing modalities and heterogeneous relative performance monitoring various time-varying environmental phenomena, and phoretic interactions enabling some robots to access otherwise unreachable regions with assistance from others.

The remainder of this paper is organized as follows. Closely related work and their differences from this paper are reviewed in Section II, followed by the necessary notation and definitions presented in Section III. In Section IV, the heterogeneous prey dynamics is constructed, which encodes the dynamic environmental conditions in the context of agent-environment interactions. In Section V, the heterogeneous predator dynamics is derived, followed by the collaborative pursuit framework that incorporates the coupled heterogeneous prey and predator dynamics. In Section VI, four different applications are presented, which demonstrate the adaptivity and effectiveness of the proposed framework, while Section VII concludes the paper.

II. RELATED WORK

In this section, we review several closely related studies and discuss the advantages of the proposed general collaborative pursuit framework in certain aspects compared with the approaches that are specifically designed for individual tasks.

On coverage control over time-varying densities, the algorithm proposed in [18] is premised on homogeneous multi-robot systems in the sense that the densities are equally accessible to all robots and the distance metric of each robot for spatial tessellations is the same leading to the standard Voronoi diagram, e.g., [19]. Different densities accessible to different robots are discussed in [20], where the spatial tessellations are also based on the standard Voronoi diagram. To encode heterogeneous relative performance or characteristics among different robots, the multiplicatively weighted Voronoi diagram is utilized in [17], where, however, the density distributions have to be equally accessible to all robots, and the algorithmic decentralization requires an approximation based on Neumann series. Improved over the aforementioned works [17], [18], [20], the coverage control strategy proposed in this paper enables heterogeneous robots to optimally cover different time-varying densities accessible to different robots in a decentralized manner without Neumann series approximation, based on the multiplicatively weighted Voronoi diagram encoding different relative performance or characteristics among robots. The proposed coverage control strategy is particularly demonstrated in Section VI-C and Section VI-D.

On pursuit–evasion problems, a typical approach is to solve the Hamilton–Jacobi–Isaacs (HJI) equations corresponding to differential games [21]. However, the HJI-based approach does not scale well because of high computational demands. A scalable pursuit–evasion scheme based on Voronoi tessellations is proposed in [22], which is suitable for the situation with one evader. A decentralized Voronoi-tessellation-based scheme for multiple pursuers to capture multiple evaders is proposed in [23], where each pursuer moves to the centroid of the bisector between the pursuer and its nearest evader. However, [23] assumes that all pursuers and evaders have the same maximum speed, which would require additional mechanisms when applying the scheme to scenarios involving heterogeneous agents. Unlike the aforementioned works that model the evaders (prey) as individual agents, the proposed collaborative pursuit framework introduces a new scheme for the pursuit–evasion (predator–prey) problem that models the prey as spatiotemporal densities, which can manage heterogeneous scenarios effectively. This scheme is particularly demonstrated in Section VI-B.

A similar idea on the prey dynamics is presented in [17], where, however, it lacks formal justifications. In contrast, in this paper, we rigorously derive a significantly more sophisticated prey dynamics and prove its asymptotic stability under certain conditions. For the example application of multi-robot firefighting, which is demonstrated in Section VI-A, the processes such as fuel combustion and thermal advection considered in this paper are not incorporated in [17]. Moreover, the prey dynamics in [17] does not incorporate the intrinsic drift process of the prey and the influence of the predators’ positions and velocities on the prey, while we consider both of them in this paper. In addition, the method in [17] is unable to handle various prey species while our proposed method is capable of doing so.

On multi-robot collaboration leveraging complementary ca-

pabilities, an aerial-aquatic robotic collaboration strategy is developed in [24] for 3D reconstruction of partially submerged targets. An amphibious-ground robotic collaboration strategy is proposed in [25] for efficient area coverage, and an aerial-ground robotic collaboration strategy is proposed in [26] for multi-target tracking. In addition to the collaboration mechanisms considered in the aforementioned works [24]–[26], where different robots perform subtasks they excel at or are uniquely capable of to improve task performance, we introduce an approach, resulting from the proposed collaborative pursuit framework, that also facilitates a more sophisticated collaboration mechanism. This mechanism enables robots to enhance their capabilities through collaborations to accomplish certain tasks that they would otherwise be unable to achieve, which is demonstrated in Section VI-D.

III. DEFINITIONS

Since the dynamics of a large class of robotic systems modeled using Euler-Lagrange equations, including aquatic, ground, and aerial robots, can be written in a control affine form, e.g., [27], [28], we consider the dynamical system

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) + G(\mathbf{x})\mathbf{u}, \quad (1)$$

where $\mathbf{x} \in \mathcal{X} \subset \mathbb{R}^p$, $\mathbf{u} \in \mathcal{U} \subset \mathbb{R}^q$, in which $p, q \in \mathbb{Z}^+$, and the locally Lipschitz vector fields $\mathbf{f} : \mathcal{X} \rightarrow \mathbb{R}^p$ and $G : \mathcal{X} \rightarrow \mathbb{R}^{p \times q}$ represent the dynamics without the control input and the influence of the control input on the system, respectively.

We denote the convex and bounded domain of interest by $\mathcal{D} \subset \mathbb{R}^d$ in which $d \in \mathbb{Z}^+$, the boundary of $\mathcal{D}$ by $\partial\mathcal{D}$, and an arbitrary point in $\mathcal{D}$ by $\mathbf{q} \in \mathcal{D}$. We let the set of indices of prey species be $\mathcal{M} = \{1, 2, \dots, M\}$, the set of indices of predators be $\mathcal{N} = \{1, 2, \dots, N\}$, where $M, N \in \mathbb{Z}^+$, the position of Predator i be $\mathbf{p}_i \in \mathcal{D}$, $\forall i \in \mathcal{N}$, and the positions of all predators be $\mathbf{p} = [\mathbf{p}_1^\top, \mathbf{p}_2^\top, \dots, \mathbf{p}_N^\top]^\top \in \mathcal{D}^N \subset \mathbb{R}^{Nd}$. Without additional specification, we typically use i, j, k to denote elements in $\mathcal{N}$ and m, n to denote elements in $\mathcal{M}$. Additionally, we denote the closed Euclidean ball with the center $\mathbf{o} \in \mathbb{R}^d$ and radius $r \in \mathbb{R}_{>0}$ by $\mathcal{B}(\mathbf{o}, r) = \{\mathbf{q} \in \mathbb{R}^d \mid \|\mathbf{q} - \mathbf{o}\| \leq r\}$, and the inner product between two d -dimensional vectors $\mathbf{a}$ and $\mathbf{b}$ by $\langle \mathbf{a}, \mathbf{b} \rangle = \sum_{z=1}^d a_z b_z$.

The Boolean operations “AND” and “OR” are denoted by $\wedge$ and $\vee$ respectively. A function belonging to C^k means that its first k derivatives exist and are continuous over the domain of the corresponding argument, and a function is said to be smooth with respect to its argument if it belongs to C^∞ . A zero vector is denoted by $\mathbf{0}_d \in \mathbb{R}^d$, a zero matrix is denoted by $\mathbf{0}_d \in \mathbb{R}^{d \times d}$, and an identity matrix is denoted by $I_d \in \mathbb{R}^{d \times d}$.

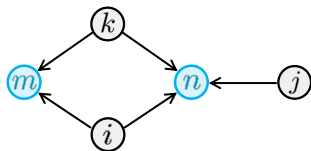


Figure 1. A predator-prey graph, where Prey Species n is pursued (targeted) by Predators i, j , and k , and Prey Species m is pursued (targeted) by Predators i and k .

Throughout the paper, we sometimes omit the subscript of the dimension indicator d for notational simplicity.

As discussed in Section I, agent-environment interactions can be described in the context of predator-prey interactions. To that end, we define the predator-prey graph, as illustrated in Fig. 1, to encode which predator (agent or robot) pursues (targets or interacts with) which prey species (types of environmental condition(s)).

Definition III.1. *The predator-prey graph is the directed graph, $\mathcal{G}_{\text{prey}}^{\text{pred}} = (\mathcal{V}_{\text{prey}}^{\text{pred}}, \mathcal{E}_{\text{prey}}^{\text{pred}})$, that specifies which predators pursue which prey species, in the sense that $(i, n) \in \mathcal{E}_{\text{prey}}^{\text{pred}}$ if and only if Predator $i \in \mathcal{N}$ pursues Prey Species $n \in \mathcal{M}$, where $i, n \in \mathcal{V}_{\text{prey}}^{\text{pred}}$.*

The idea of different predators pursuing different prey species encoded by the predator-prey graph can be interpreted as different task suitability among heterogeneous robots, illustrating how a group of agents should collaboratively accomplish certain tasks that are challenging for individual agents.

We denote the out-neighborhood of Predator i as $\mathcal{M}_i^{\mathcal{N}} = \{n \in \mathcal{M} \mid (i, n) \in \mathcal{E}_{\text{prey}}^{\text{pred}}\}$, i.e., the set of indices of all prey species that are pursued by Predator i , and the in-neighborhood of Prey Species n as $\mathcal{N}_n^{\mathcal{M}} = \{i \in \mathcal{N} \mid (i, n) \in \mathcal{E}_{\text{prey}}^{\text{pred}}\}$, i.e., the set of indices of all predators that pursue Prey Species n .

Note that the terms *predator*, *agent*, and *robot* are used interchangeably to denote the same concept in this paper. In addition, there is no ambiguity when the same number is used to label the vertex of $\mathcal{G}_{\text{prey}}^{\text{pred}}$ representing prey and the one representing predator since the in-degree of each predator vertex and the out-degree of each prey vertex are always zero per Definition III.1.

IV. HETEROGENEOUS PREY DYNAMICS

Inspired by a type of predator-prey interactions in nature, such as whales and krill, e.g., [29], herbivores and vegetation, e.g., [30], and sharks and fish schools, e.g., [31], we model the prey as continuous densities while the predators as discrete individual agents. Modeling the prey as density distributions offers advantages. For example, when the prey population is large, keeping track of each individual is challenging, while analyzing the dynamics of the ensemble is both analytically and computationally efficient. This is reminiscent of the Eulerian description compared with the Lagrangian description in the context of fluid dynamics, and the macroscopic description compared with the microscopic description in the context of statistical physics. In addition, the motion of each individual prey, governed by a certain principle, gives rise to collective behavior of the assemblage of prey, setting the stage for analysis of the behavior of prey (either with or without the influence of the predators) through PDEs.

Let $\Omega \subseteq \mathcal{D}$ be an arbitrary control volume and denote the boundary of Ω as $\partial\Omega$. For Prey Species $n \in \mathcal{M}$, we let its concentration, i.e., the amount per unit volume, be denoted as the space-time-dependent density $\phi_n : \mathcal{D} \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$, which is assumed to belong to $C^2(\mathbf{q})$ and $C^1(t)$. In addition, we assume the Lebesgue measure $\mu(o(t)) = 0, \forall o(t) \in 2^{\mathcal{O}(t)}, \forall t \in \mathbb{R}_{\geq 0}$, where $\mathcal{O}(t) = \{\mathbf{q} \in \mathcal{D} \mid \phi_n(\mathbf{q}, t) = 0\}$, and $2^{\mathcal{O}(t)}$

denotes the power set of $\mathcal{O}(t)$, such that the predator dynamics (discussed in detail in Section V) is well defined.

As described in [32], in a prey assemblage, each individual prey (of one species) can be modeled as moving at a microscopic scale following a simple bordered symmetric random walk without the influence of predators and other prey species. Thus, a diffusion process of the assemblage emerges at a macroscopic scale. In addition to the random walk, we assume that each individual prey also migrates according to the characteristics of its species, from which a drift process of the assemblage emerges. To capture both diffusion and drift processes of a physical quantity, we define the flux of Prey Species $n \in \mathcal{M}$, $\mathbf{J}_n : \mathcal{D} \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^d$, which is assumed to be a C^1 vector field on a neighborhood of Ω and represents the amount of the prey moving through a unit area on $\partial\Omega$ per unit time t , to be

$$\mathbf{J}_n(\mathbf{q}, t) = \mathbf{\Gamma}_n(\mathbf{q}, t)\phi_n(\mathbf{q}, t) - D_n(\mathbf{q}, t)\nabla_{\mathbf{q}}\phi_n(\mathbf{q}, t), \quad (2)$$

in which $\nabla_{\mathbf{q}}$ denotes the Nabla operator with respect to $\mathbf{q}$, $D_n : \mathcal{D} \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ denotes the diffusivity of the diffusion process of Prey Species n , and $\mathbf{\Gamma}_n : \mathcal{D} \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^d$ denotes the vector field governing the intrinsic drift process. Note that the term *intrinsic* is used to describe the process without the influence of predators.

Since multiple prey species can be involved and different species may interact with each other's diffusion process according to [33], we consider cross-diffusivity $D_{nm} : \mathcal{D} \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$, $\forall m, n \in \mathcal{M}$, to encode the mutual interactions of diffusion processes among different prey species. Then, (2) can be rewritten as $\mathbf{J}_n(\mathbf{q}, t) = \mathbf{\Gamma}_n(\mathbf{q}, t)\phi_n(\mathbf{q}, t) - \sum_{m \in \mathcal{M}} [D_{nm}(\mathbf{q}, t)\nabla_{\mathbf{q}}\phi_m(\mathbf{q}, t)]$, where we denote $D_{nn} = D_n$ to make the expression compact. Note that the cross-diffusion process does not arise often in practical biological models, as discussed in [33], so we typically assume $D_{nm} \ll D_{nn}$, $\forall m \neq n \in \mathcal{M}$, in this paper.

With the introduction of predators, the intrinsic dynamics of prey will be affected accordingly. We consider the effects of pursuit and evasion, i.e., when the predators are pursuing the prey, the prey may be evading the predators, and thus the vector field governing the intrinsic drift process of the ensemble of prey can be augmented by the positions and velocities of the predators. As such, we define the augmented vector field $\tilde{\mathbf{\Gamma}}_n : \mathcal{D} \times \mathbb{R}_{\geq 0} \times \mathcal{D}^N \times \mathbb{R}^{Nd} \rightarrow \mathbb{R}^d$ under the influence of predators as

$$\tilde{\mathbf{\Gamma}}_n(\mathbf{q}, t, \mathbf{p}, \dot{\mathbf{p}}) = \mathbf{\Gamma}_n(\mathbf{q}, t) + \sum_{i \in \mathcal{N}_{\mathcal{M}}^n} \Upsilon_{n,i}(\mathbf{q}, \mathbf{p}_i, \dot{\mathbf{p}}_i), \quad (3)$$

where

$$\Upsilon_{n,i}(\mathbf{q}, \mathbf{p}_i, \dot{\mathbf{p}}_i) = \exp\left(-\frac{\|\mathbf{p}_i - \mathbf{q}\|^2}{\chi_{n,i}^2}\right) \Lambda_{n,i}\dot{\mathbf{p}}_i(t) \quad (4)$$

encodes the influence of Predator i on Prey Species n 's intrinsic drift process such that the influence is stronger when closer to the predator, in which $\chi_{n,i} \in \mathbb{R}_{>0}$ characterizes the size of Predator i 's influence zone, and $\Lambda_{n,i} \in \mathbb{R}^{d \times d}$ characterizes the strength and direction of the influence of Predator i 's velocity on Prey Species n 's intrinsic drift process, which is determined

by the specifics of the predator-prey interaction scheme. Then, the augmented flux of Prey Species n under the influence of predators can be written as

$$\tilde{\mathbf{J}}_n(\mathbf{q}, t, \mathbf{p}, \dot{\mathbf{p}}) = \tilde{\mathbf{\Gamma}}_n(\mathbf{q}, t, \mathbf{p}, \dot{\mathbf{p}})\phi_n(\mathbf{q}, t) - \sum_{m \in \mathcal{M}} [D_{nm}(\mathbf{q}, t)\nabla_{\mathbf{q}}\phi_m(\mathbf{q}, t)]. \quad (5)$$

Since vegetation, regarded as the prey of herbivores, can sprout and wither, and krill, as the prey of whales, can reproduce and perish, we define $R_n : \mathcal{D} \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ as the reaction process encoding how Prey Species n intrinsically emerges or disappears. Furthermore, similar to (3), the augmented reaction process of Prey Species n under the influence of predators is defined as

$$\tilde{R}_n(\mathbf{q}, t, \mathbf{p}) = R_n(\mathbf{q}, t) + \sum_{i \in \mathcal{N}_{\mathcal{M}}^n} \Psi_{n,i}(\mathbf{q}, t, \mathbf{p}_i), \quad (6)$$

where

$$\Psi_{n,i}(\mathbf{q}, t, \mathbf{p}_i) = \begin{cases} f_{n,i}^R(\mathbf{q}, t, \mathbf{p}_i) & \text{if } \mathbf{q} \in \mathcal{B}(\mathbf{p}_i, r_{n,i}^R) \\ 0 & \text{if } \mathbf{q} \notin \mathcal{B}(\mathbf{p}_i, r_{n,i}^R + \delta_{n,i}^R) \\ g_{n,i}^R(\mathbf{q}, t, \mathbf{p}_i) & \text{otherwise} \end{cases} \quad (7)$$

represents the reaction component caused by the influence of Predator i on Prey Species n , where $r_{n,i}^R$ is the radius of Predator i 's influence zone on Prey Species n 's reaction processes, $f_{n,i}^R : \mathcal{D} \times \mathbb{R}_{\geq 0} \times \mathbb{R}^d \rightarrow \mathbb{R}$ is locally Lipschitz with respect to $\mathbf{q}$, which should be determined by specific predator-prey interaction scheme, and

$$g_{n,i}^R(\mathbf{q}, t, \mathbf{p}_i) = \frac{f_{n,i}^R(\mathbf{q}, t, \mathbf{p}_i)}{\delta_{n,i}^R} (\delta_{n,i}^R + r_{n,i}^R - \|\mathbf{p}_i - \mathbf{q}\|), \quad (8)$$

in which $\delta_{n,i}^R \ll r_{n,i}^R$ serves as an interpolation interval between $f_{n,i}^R$ and $g_{n,i}^R$ to ensure that $\Psi_{n,i} \in C^0(\mathbf{q})$, which is essential for Proposition IV.1. Accordingly, the properties of the reaction process of Prey Species n affected by Predator i is determined by $f_{n,i}^R(\mathbf{q}, t, \mathbf{p}_i)$.

In order to derive prey dynamics (and also predator dynamics which will be discussed in detail in Section V), we leverage Leibniz-Reynolds transport theorem [34], presented in Lemma IV.1.

Lemma IV.1. *Given the real-valued function*

$$L(\mathbf{p}, t) = \int_{\Omega} \ell(\mathbf{p}, t, \mathbf{q}) d\mathbf{q},$$

if $\ell : \mathbb{R}^{Nd} \times \mathbb{R}_{\geq 0} \times \mathcal{D} \rightarrow \mathbb{R}$ belongs to C^1 with respect to all its arguments, and $\Omega \subseteq \mathcal{D} \subset \mathbb{R}^d$ depends smoothly on $\mathbf{p}$ such that the outward-pointing unit normal $\hat{\mathbf{n}}$ is uniquely defined almost everywhere on $\partial\Omega$, $\forall t \in \mathbb{R}_{\geq 0}$, then

$$\frac{\partial}{\partial \mathbf{s}} L(\mathbf{p}, t) = \int_{\Omega} \frac{\partial}{\partial \mathbf{s}} \ell(\mathbf{p}, t, \mathbf{q}) d\mathbf{q} + \int_{\partial\Omega} \ell(\mathbf{p}, t, \mathbf{q}) \hat{\mathbf{n}}^\top \frac{\partial}{\partial \mathbf{s}} \mathbf{q} d\mathbf{q}.$$

in which $\mathbf{s}$ can be $\mathbf{p}$, $\mathbf{p}_i$, or t .

Proof. See [35]. $\square$

Primarily using Lemma IV.1, we obtain the prey dynamics in Proposition IV.1.

Proposition IV.1. If $\phi_n(\mathbf{q}, t) \in C^2(\mathbf{q})$, $\phi_n(\mathbf{q}, t) \in C^1(t)$, $\Gamma_n(\mathbf{q}, t) \in C^1(\mathbf{q})$, $D_{nm}(\mathbf{q}, t) \in C^1(\mathbf{q})$, $R_n(\mathbf{q}, t) \in C^0(\mathbf{q})$, and $f_{n,i}^R(\mathbf{q}, t, \mathbf{p}_i) \in C^0(\mathbf{q})$, $\forall m, n \in \mathcal{M}$, $\forall i \in \mathcal{N}$, then the heterogeneous prey dynamics is given by the augmented Fokker–Planck equation

$$\boxed{\frac{\partial}{\partial t} \phi + \nabla_{\mathbf{q}} \circ (\text{diag}(\phi) \tilde{\Gamma} - \tilde{D} \nabla_{\mathbf{q}} \phi) - \tilde{\mathbf{R}} = \mathbf{0}} \quad (9)$$

in which $\phi = [\phi_1, \phi_2, \dots, \phi_M]^\top \in \mathbb{R}^M$ denotes the vector containing the densities of the M prey species, the operator $\circ$ represents a row-wise inner product such that $\nabla_{\mathbf{q}} \circ A := [\langle \nabla_{\mathbf{q}}, \mathbf{a}_1 \rangle, \langle \nabla_{\mathbf{q}}, \mathbf{a}_2 \rangle, \dots, \langle \nabla_{\mathbf{q}}, \mathbf{a}_M \rangle]^\top$ where $A = [\mathbf{a}_1^\top, \mathbf{a}_2^\top, \dots, \mathbf{a}_M^\top]^\top \in \mathbb{R}^{M \times d}$, $\text{diag}(\phi) = \text{diag}(\phi_1, \phi_2, \dots, \phi_M) \in \mathbb{R}^{M \times M}$ denotes a diagonal matrix, $\tilde{\Gamma} = [\tilde{\Gamma}_1, \tilde{\Gamma}_2, \dots, \tilde{\Gamma}_M]^\top \in \mathbb{R}^{M \times d}$ denotes the augmented drift matrix, $\tilde{D} \in \mathbb{R}^{M \times M}$ denotes the diffusion matrix whose entries $[\tilde{D}]_{nm} = D_{nm}$, and $\tilde{\mathbf{R}} = [\tilde{R}_1, \tilde{R}_2, \dots, \tilde{R}_M]^\top \in \mathbb{R}^M$ denotes the augmented reaction vector. The arguments of the quantities in (9) are omitted for concision of the expression.

Proof. According to the fundamental conservation law of nature, i.e., the rate of change of the amount of certain conserved quantity, e.g., mass, energy, matter, in total in Ω with respect to time is equal to the sum of the flux of the quantity through the boundary $\partial\Omega$ and the quantity generated or consumed inside Ω . In this paper, the conserved quantity is represented by the prey density, so for Prey Species $n \in \mathcal{M}$, the conservation law can be expressed as

$$\begin{aligned} \frac{\partial}{\partial t} \int_{\Omega} \phi_n(\mathbf{q}, t) d\mathbf{q} = & - \oint_{\partial\Omega} \langle \hat{\mathbf{n}}, \tilde{\mathbf{J}}_n(\mathbf{q}, t, \mathbf{p}, \dot{\mathbf{p}}) \rangle d\mathbf{q} \\ & + \int_{\Omega} \tilde{R}_n(\mathbf{q}, t, \mathbf{p}) d\mathbf{q}, \end{aligned} \quad (10)$$

where $\hat{\mathbf{n}}$ denotes the outward-pointing unit normal on $\partial\Omega$.

According to the Gauss's divergence theorem, we have

$$\oint_{\partial\Omega} \langle \hat{\mathbf{n}}, \tilde{\mathbf{J}}_n(\mathbf{q}, t, \mathbf{p}, \dot{\mathbf{p}}) \rangle d\mathbf{q} = \int_{\Omega} \langle \nabla_{\mathbf{q}}, \tilde{\mathbf{J}}_n^\top(\mathbf{q}, t, \mathbf{p}, \dot{\mathbf{p}}) \rangle d\mathbf{q}, \quad (11)$$

where $\tilde{\mathbf{J}}_n(\mathbf{q}, t, \mathbf{p}, \dot{\mathbf{p}}) \in \mathbb{R}^d$ is a C^1 vector field defined on a neighborhood of Ω .

Based on (5) and (11), using Lemma IV.1, then (10) can be calculated as

$$\begin{aligned} \int_{\Omega} \left(\frac{\partial}{\partial t} \phi_n(\mathbf{q}, t) + \langle \nabla_{\mathbf{q}}, \tilde{\mathbf{J}}_n^\top(\mathbf{q}, t, \mathbf{p}, \dot{\mathbf{p}}) \rangle \phi_n(\mathbf{q}, t) \right. \\ \left. - \sum_{m \in \mathcal{M}} [D_{nm}(\mathbf{q}, t) \nabla_{\mathbf{q}} \phi_m(\mathbf{q}, t)] - \tilde{R}_n(\mathbf{q}, t, \mathbf{p}) \right) d\mathbf{q} = 0. \end{aligned} \quad (12)$$

If $\Gamma_n(\mathbf{q}, t) \in C^1(\mathbf{q})$, according to (3) and (4), we have that $\tilde{\Gamma}_n(\mathbf{q}, t, \mathbf{p}, \dot{\mathbf{p}}) \in C^1(\mathbf{q})$. Furthermore, if $R_n(\mathbf{q}, t) \in C^0(\mathbf{q})$ and $f_{n,i}^R(\mathbf{q}, t, \mathbf{p}_i) \in C^0(\mathbf{q})$, according to (6), (7), and (8), we have that $\tilde{R}_n(\mathbf{q}, t, \mathbf{p}) \in C^0(\mathbf{q})$. Note that the definition of $g_{n,i}^R(\mathbf{q}, t, \mathbf{p}_i)$ in (8) ensures that $\Psi_{n,i}(\mathbf{q}, t, \mathbf{p}_i) \in C^0(\mathbf{q})$ when $f_{n,i}^R(\mathbf{q}, t, \mathbf{p}_i) \in C^0(\mathbf{q})$. Therefore, if $D_{nm}(\mathbf{q}, t) \in C^1(\mathbf{q})$ for all $m, n \in \mathcal{M}$, with the assumption that $\phi_n(\mathbf{q}, t) \in C^2(\mathbf{q})$ and $\phi_n(\mathbf{q}, t) \in C^1(t)$, we have $\mathcal{J}(\mathbf{q}) \in C^0(\mathbf{q})$ where $\mathcal{J}(\mathbf{q})$ is the entire integrand in (12) with respect to $\mathbf{q}$. Since the

control volume Ω is arbitrary, then $\int_{\Omega} \mathcal{J}(\mathbf{q}) d\mathbf{q} = 0$ implies $\mathcal{J}(\mathbf{q}) \equiv 0$, which can be proved by contradiction, e.g., [16].

Therefore,

$$\begin{aligned} \frac{\partial}{\partial t} \phi_n(\mathbf{q}, t) + \langle \nabla_{\mathbf{q}}, \tilde{\mathbf{J}}_n^\top(\mathbf{q}, t, \mathbf{p}, \dot{\mathbf{p}}) \rangle \phi_n(\mathbf{q}, t) \\ - \sum_{m \in \mathcal{M}} [D_{nm}(\mathbf{q}, t) \nabla_{\mathbf{q}} \phi_m(\mathbf{q}, t)] - \tilde{R}_n(\mathbf{q}, t, \mathbf{p}) = 0. \end{aligned} \quad (13)$$

Stacking (13), $\forall n \in \mathcal{M}$, together yields the formula (9). $\square$

The asymptotic stability of the prey density under certain conditions is presented in Proposition IV.2.

Proposition IV.2. For $\phi_n : \Omega \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$, where $\Omega \subseteq \mathcal{D}$ and $n \in \mathcal{M}$, if $\forall t \in \mathbb{R}_{\geq 0}$, $\tilde{\mathbf{J}}_n(\mathbf{q}, t, \mathbf{p}, \dot{\mathbf{p}}) = 0$, $\forall \mathbf{q} \in \partial\Omega$, and $\int_{\Omega} \tilde{R}_n(\mathbf{q}, t, \mathbf{p}) d\mathbf{q} \in \mathbb{R}_{<0}$, then

$$\lim_{t \rightarrow \infty} \int_{\Omega} \phi_n(\mathbf{q}, t) d\mathbf{q} = 0.$$

Proof. Denote $P_n(t) = \int_{\Omega} \phi_n(\mathbf{q}, t) d\mathbf{q}$. Since $\phi_n(\mathbf{q}, t) \in \mathbb{R}_{\geq 0}$ and by assumption the Lebesgue measure $\mu(o(t)) = 0$, $\forall o(t) \in 2^{\mathcal{O}(t)}$ with $\mathcal{O}(t) = \{\mathbf{q} \in \mathcal{D} \mid \phi_n(\mathbf{q}, t) = 0\}$, $\forall t \in \mathbb{R}_{\geq 0}$, then

$$P_n(t) \in \mathbb{R}_{>0}, \forall t \in \mathbb{R}_{\geq 0}.$$

According to (10), we have

$$\dot{P}_n(t) = - \oint_{\partial\Omega} \langle \hat{\mathbf{n}}, \tilde{\mathbf{J}}_n(\mathbf{q}, t, \mathbf{p}, \dot{\mathbf{p}}) \rangle d\mathbf{q} + \int_{\Omega} \tilde{R}_n(\mathbf{q}, t, \mathbf{p}) d\mathbf{q}.$$

Since $\tilde{\mathbf{J}}_n(\mathbf{q}, t, \mathbf{p}, \dot{\mathbf{p}}) = 0$, $\forall \mathbf{q} \in \partial\Omega$, $t \in \mathbb{R}_{\geq 0}$, then

$$\oint_{\partial\Omega} \langle \hat{\mathbf{n}}, \tilde{\mathbf{J}}_n(\mathbf{q}, t, \mathbf{p}, \dot{\mathbf{p}}) \rangle d\mathbf{q} = 0.$$

Additionally, since $\int_{\Omega} \tilde{R}_n(\mathbf{q}, t, \mathbf{p}) d\mathbf{q} \in \mathbb{R}_{<0}$, $\forall t \in \mathbb{R}_{\geq 0}$, then

$$\dot{P}_n(t) \in \mathbb{R}_{<0}, \forall t \in \mathbb{R}_{\geq 0}.$$

Therefore,

$$\lim_{t \rightarrow \infty} P_n(t) = \lim_{t \rightarrow \infty} \int_{\Omega} \phi_n(\mathbf{q}, t) d\mathbf{q} = 0.$$

$\square$

An intuition of Proposition IV.2 is that if there is no flux of Prey Species n on $\partial\Omega$ for all time, and its net reaction process across Ω has a “consuming” effect for all time, then Prey Species n will be “consumed” asymptotically.

Since the intrinsic processes of prey such as drift and reaction can also be affected by their ambient environmental factors, we explicitly define the hierarchical environmental conditions mentioned in Section I as follows. The ones that directly interact with the predators (agents or robots) are only referred to as *prey*, whereas the ones that affect the dynamics of the prey but do not directly affect the predators are only referred to as *environmental factors*, and the term *environmental conditions* can refer to either of them, as illustrated in Fig. 2. For example, in the context of forest firefighting, the evolving forest fire that directly interacts with the firefighting robots is referred to as prey, the vegetation (fuel) that affects the dynamics of the forest fire instead of directly affecting the dynamics of the firefighting robots is

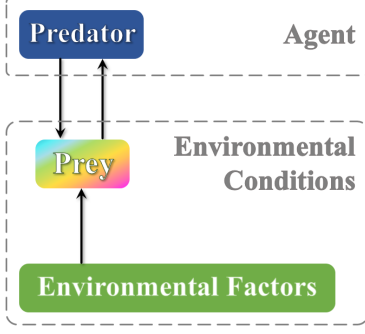


Figure 2. Illustration of the concepts of predator (agent or robot), prey, environmental factors, and environmental conditions defined in this paper.

referred to as environmental factor, and either the forest fire or the vegetation can be referred to as environmental condition of the firefighting robots.

Note that drift, diffusion, reaction process, and influence of predators are expressed in (9) to concisely articulate these potential effects, which does not imply that all of them will manifest simultaneously in every specific application scenario. For example, regarding herbivores foraging on vegetation, the vegetation density does not exhibit drift and diffusion.

To summarize the purpose of each component of the prey dynamics (9), the eight heterogeneous modalities (that are user-defined according to specific tasks) incorporated in (9) and their corresponding descriptions are listed in Table I, where we also include some types of heterogeneity related to predators because they are coupled with the heterogeneous prey dynamics (9) and difficult to separate.

To show how key components (e.g., drift, diffusion, and reaction) of the prey dynamics (9) function, we present the following simple example use cases¹ with one predator and one prey species in $\mathcal{D} \subset \mathbb{R}^2$, illustrating the effects of corresponding parameters in (9). More complex cases incorporating predator dynamics, developed in Section V, will be demonstrated in Section VI.

First, we demonstrate how the reaction and diffusion terms function without the drift term. Let $N = M = 1$ and $\tilde{\Gamma}_1 = \mathbf{0}$, then (9) degenerates into $\frac{\partial}{\partial t} \phi_1 + \langle \nabla_{\mathbf{q}}, D_{11} \nabla_{\mathbf{q}} \phi_1 \rangle - \tilde{R}_1 = 0$, which is a reaction-diffusion equation, where the arguments are omitted for concision in expression. Let $R_1 = 0$, $\delta_{1,1}^R = 10^{-3}$, and $f_{1,1}^R = -500 \cdot \phi_1$. The simulation results, with different diffusivities D_{11} and radii of the predator's influence zone on the prey's reaction process $r_{1,1}^R$, are shown in Fig. 3, from which one can observe that the prey density diffuses faster with a larger diffusivity, and the predator has a wider "consuming" range due to a larger radius of the corresponding influence zone.

Second, we demonstrate how the drift and diffusion terms function without the reaction term. Let $N = M = 1$ and $\tilde{R}_1 = 0$, then (9) degenerates into $\frac{\partial}{\partial t} \phi_1 + \langle \nabla_{\mathbf{q}}, \tilde{\Gamma}_1^\top \phi_1 - D_{11} \nabla_{\mathbf{q}} \phi_1 \rangle = 0$, which is an augmented drift-diffusion equation. Regarding fixed parameters, we let $D_{11} = 1$, and $\Gamma_1(\mathbf{q}) = \begin{bmatrix} 0 & -80 \\ 40 & 0 \end{bmatrix} \cdot \mathbf{q}$.

¹A video of the full simulations corresponding to Fig. 3 and Fig. 4 is available in the supplementary multimedia files.

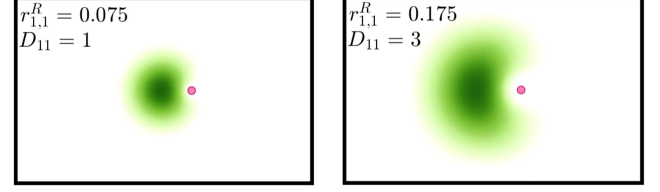


Figure 3. Snapshots of the simulation results of $\frac{\partial}{\partial t} \phi_1 + \langle \nabla_{\mathbf{q}}, D_{11} \nabla_{\mathbf{q}} \phi_1 \rangle - \tilde{R}_1 = 0$ at the same final time step under different parameter settings. The magenta dot represents the stationary predator, and the green density distribution represents the prey with darker-colored areas indicating higher density while lighter-colored areas indicating lower density.

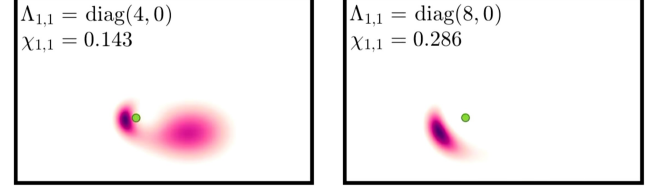


Figure 4. Snapshots of the simulation results of $\frac{\partial}{\partial t} \phi_1 + \langle \nabla_{\mathbf{q}}, \tilde{\Gamma}_1^\top \phi_1 - D_{11} \nabla_{\mathbf{q}} \phi_1 \rangle = 0$ at the same final time step under different parameter settings. The green dot represents the predator in uniform linear motion to the left, and the magenta density distribution represents the prey with darker-colored areas indicating higher density while lighter-colored areas indicating lower density.

The simulation results, with different sizes of the predator's influence zone $\chi_{1,1}$ and magnitudes of the influence of the predator's velocity on the prey's drift process $\Lambda_{1,1}$, are shown in Fig. 4. It can be observed from Fig. 4 that the weaker the influence of the predator on the prey, i.e., the smaller $\Lambda_{1,1}$ and $\chi_{1,1}$, the more easily the prey follows its intrinsic drift pattern.

Note that how the intrinsic drift and reaction processes of prey are affected by predators are defined by (3) and (6) without loss of generality. They can furthermore be modified based on specific application scenarios or task requirements as long as the conditions in Proposition IV.1 such that (9) holds

Table I
HETEROGENEOUS MODALITIES INCORPORATED IN (9)

Symbol	Description
D_{nn}	Intrinsic diffusion rate of Prey Species n
D_{nm}	Influence of Prey Species m on the diffusion of Prey Species n , typically much smaller than D_{nn}
Γ_n	The vector field governing the intrinsic drift of Prey Species n
$\chi_{n,i}$	Size of the influence zone of Predator i on the drift of Prey Species n
$\Lambda_{n,i}$	Influence of the velocity of Predator i on the magnitude and direction of the drift of Prey Species n
R_n	Intrinsic reaction process of Prey Species n , encoding its natural emergence and disappearance
$r_{n,i}^R$	Radius of the influence zone of Predator i on the reaction process of Prey Species n
$f_{n,i}^R$	Influence of Predator i on the reaction process of Prey Species n

are satisfied. In addition, although we model the prey dynamics from a biological point of view as an inspiration, as discussed in Section I, what the prey density together with its evolution governed by (9) can encode is general. For example, thermal energy to be consumed, information to be gathered, ensemble of evaders to be pursued, environmental phenomena to be monitored, and user-specified principles intended to guide the behavior of robots.

The heterogeneous nature of (9) enables the specification of various combinations of the heterogeneous modalities listed in Table I, which in turn, along with the heterogeneous predator dynamics introduced in the next section, allows for addressing diverse tasks, as will be demonstrated in Section VI in detail.

V. HETEROGENEOUS PREDATOR DYNAMICS

The prey dynamics developed in Section IV encodes changing environmental conditions with or without the influence of predators (agents). In turn, the prey dynamics can also influence predator dynamics. To encode the predator-prey (agent-environment) interaction, one can design the predator dynamics for collaborative pursuit in the way that each predator acts based on its characteristics and local environment, and collaborates with its neighbors, resulting in a collective behavior of the entire group of predators susceptible to the prey dynamics. To this end, we make three assumptions about the predator dynamics:

- (a) Since the target prey species can vary among different types of predators in the domain of interest, to promote spatial divisions of labor, inspired by the behavior of territorial animals, e.g., [15], [36], each predator is assumed to be in charge of a dominant subdomain with respect to each of its target prey species. The pattern of each dominant subdomain is determined by the corresponding predator's relative performance or characteristic among other predators;
- (b) Each predator acts only according to the local information of its topologically neighboring (inspired by the study in [37]) predators and the prey within its dominant subdomain(s); and
- (c) The prey are optimally covered by corresponding predators so that there exist more predators in an area with higher density of prey while relatively less predators in an area with relatively lower density of prey.

Note that the dominant subdomain in Assumption (a) of each predator should be geometrically non-pathological, i.e., its Lebesgue measure is nonzero and it includes the corresponding predator within it for all time. In addition, Assumption (a) implies that a predator can have multiple dominant subdomains with respect to multiple prey species simultaneously.

Inspired by the coverage strategy introduced in [38], leveraging the idea of territoriality of robots, we define the time-varying locational cost function encoding the lack of coverage quality of all the predators over all prey species as

$$F(\mathbf{p}, t) = \sum_{n \in \mathcal{M}} \int_{\mathcal{D}} \min_{i \in \mathcal{N}_{\mathcal{M}}^n} f(d_{i,n}(\mathbf{p}_i, \mathbf{q})) \phi_n(\mathbf{q}, t) d\mathbf{q}, \quad (14)$$

in which $d_{i,n} : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}_{\geq 0}$ is the distance metric for Predator i to pursue Prey Species n at $\mathbf{q}$ defined based on certain characteristics of them. Moreover, $f(\cdot) \in C^1 : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is a monotonically increasing function of its argument, indicating that the farther Predator i is from $\mathbf{q}$, the less effective coverage of Predator i on $\mathbf{q}$.

As such, different task suitability among different robots encoded by the predator-prey graph $\mathcal{G}_{\text{prey}}^{\text{pred}}$ per Definition III.1 can be concretely characterized by having different robots cover different densities. For example, underwater robots equipped with water quality sensors can monitor (cover) chemical pollutants (densities) in specific areas, while others equipped with sonar can monitor (cover) the diffusion and drift of these pollutants (densities) over a wider range. These robots, by working together, provide a comprehensive assessment of marine pollution levels.

The cost function (14) can be rewritten as

$$F(\mathbf{p}, t) = \sum_{n \in \mathcal{M}} \sum_{i \in \mathcal{N}_{\mathcal{M}}^n} \int_{\mathcal{V}_{i,n}(\mathbf{p})} d_{i,n}^2(\mathbf{p}_i, \mathbf{q}) \phi_n(\mathbf{q}, t) d\mathbf{q}, \quad (15)$$

assuming $f(\cdot) = (\cdot)^2$, where $\mathcal{V}_{i,n}(\mathbf{p})$ is the dominant subdomain of Predator i with respect to Prey Species n . In $\mathcal{V}_{i,n}(\mathbf{p})$, Predator i is closer, in the sense of the distance metric $d_{i,n}(\cdot, \cdot)$, to any point $\mathbf{q} \in \mathcal{V}_{i,n}(\mathbf{p})$ than any other predators pursuing $\phi_n(\mathbf{q}, t)$. Specifically,

$$\mathcal{V}_{i,n}(\mathbf{p}) = \left\{ \mathbf{q} \in \mathcal{D} \mid d_{i,n}(\mathbf{p}_i, \mathbf{q}) \leq d_{j,n}(\mathbf{p}_j, \mathbf{q}), \right. \\ \left. \forall j \neq i, i, j \in \mathcal{N}_{\mathcal{M}}^n \right\}. \quad (16)$$

Inspired by [39], where the heterogeneous relative performance or characteristics in terms of each robot's mobility is encoded by a Riemannian metric, we define the distance metric $d_{i,n}(\cdot, \cdot)$ in (16) as a multiplicatively weighted distance

$$d_{i,n}(\mathbf{p}_i, \mathbf{q}) := w_{i,n} \|\mathbf{p}_i - \mathbf{q}\|, \quad (17)$$

where the weighting metric $w_{i,n} \in \mathbb{R}_{>0}$, $\forall i \in \mathcal{N}_{\mathcal{M}}^n$. Hence, (17) can encode the relative performance or characteristic of Predator i among a team of predators (determined by $\mathcal{N}_{\mathcal{M}}^n$) with respect to Prey Species n . As a result, the smaller the weighting metric $w_{i,n}$ corresponding to a predator, the relatively larger the area (under the Lebesgue measure) of the dominant subdomain of the predator when pursuing Prey Species n . Therefore, the neighboring predators of Predator i with respect to Prey Species n mentioned in Assumption (b) refer specifically to the Delaunay neighbors of Predator i in the sense of the multiplicatively weighted Voronoi diagram.

Similar to the weighting coefficient utilized in [40] to encode general heterogeneous relative performance among different robots based on the Power diagram [41], the weighting metric $w_{i,n}$ in (17) can encode general attributes, such as mobility, e.g., [42], sensing range, e.g., [26], [43], or other user-specified metrics that can comprehensively represent various types of performance or characteristics based on specific application preferences.

Although the additively weighted Voronoi diagram and the Power diagram also seemed to be able to achieve Assumption (a) considering heterogeneous relative performance, we

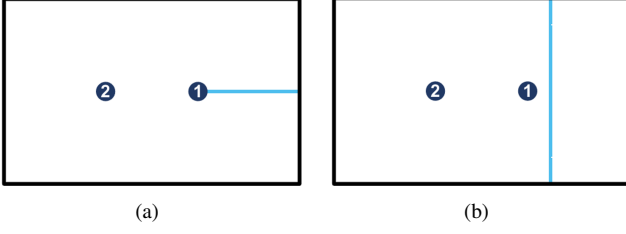


Figure 5. Examples of geometrically pathological Voronoi diagrams of two robots, where the sky-blue lines represent the bisectors of the two robots with $\mathbf{p}_1 = [0.5, 0]^\top$ and $\mathbf{p}_2 = [-0.5, 0]^\top$. (a) Additively weighted Voronoi diagram with $w_1 = 1$ and $w_2 = 2$ based on (18); (b) Power diagram with $w_1 = 1$ and $w_2 = 2.5$ based on (19).

utilize the multiplicatively weighted Voronoi diagram to define the predators' dominant subdomains. The reason for this is twofold. First, from a theoretical point of view, if we consider $\mathbb{R}^d$ as a special type of Riemannian manifold, whose tangent space is also $\mathbb{R}^d$, there does not exist a Riemannian metric such that the resulting geodesic can be

$$d(\mathbf{p}_i, \mathbf{q}) = \|\mathbf{p}_i - \mathbf{q}\| - w_i \quad (18)$$

or

$$d(\mathbf{p}_i, \mathbf{q}) = \|\mathbf{p}_i - \mathbf{q}\|^2 - w_i, \quad (19)$$

corresponding to the additively weighted Voronoi diagram and the Power diagram, respectively. Second, from a practical point of view, the additively weighted Voronoi diagram and the Power diagram cannot always be geometrically non-pathological, as shown in [19]. For instance, when (18) is applied and $\|\mathbf{p}_2(t) - \mathbf{p}_1(t)\| \leq w_2 - w_1$, the dominant subdomain of Robot 1 becomes a zero Lebesgue measure set as shown in Fig. 5(a). When (19) is applied and $\|\mathbf{p}_2(t) - \mathbf{p}_1(t)\|^2 < w_2 - w_1$, Robot 1 is located outside of its dominant subdomain as shown in Fig. 5(b). Thus, if the weights are not carefully chosen or the relations between the weights and the positions of the agents are not carefully treated, either the additively weighted Voronoi diagram or the Power diagram are often not able to achieve Assumption (a), especially in such dynamic environments considered in this paper.

In contrast, the multiplicatively weighted Voronoi diagram based on the distance metric (17) is guaranteed to be geometrically non-pathological as shown in [19] and thus achieves Assumption (a) for all time. Moreover, the analytical expression of the bisector of two robots in the multiplicatively weighted Voronoi diagram is given by Lemma V.1.

Lemma V.1. *The set of points in $\mathbb{R}^d$ that have the same multiplicatively weighted distance to $\mathbf{p}_i$ and $\mathbf{p}_j$, i.e., $w_i \|\mathbf{p}_i - \mathbf{q}\| = w_j \|\mathbf{p}_j - \mathbf{q}\|$, where $\mathbf{p}_i, \mathbf{p}_j, \mathbf{q} \in \mathbb{R}^d$, is given by the Apollonius circle, e.g., [19]*

$$\partial \mathcal{V}_{ij} = \{\mathbf{q} \in \mathbb{R}^d \mid \|\mathbf{q} - \mathbf{o}_{ij}^{\text{AC}}\| = r_{ij}^{\text{AC}}\},$$

in which

$$\mathbf{o}_{ij}^{\text{AC}} = \frac{w_i^2 \mathbf{p}_i - w_j^2 \mathbf{p}_j}{w_i^2 - w_j^2}, \text{ and } r_{ij}^{\text{AC}} = \frac{w_i w_j \|\mathbf{p}_i - \mathbf{p}_j\|}{w_i^2 - w_j^2},$$

where $w_i > w_j \in \mathbb{R}_{>0}$ without loss of generality. When $w_i = w_j$, $\partial \mathcal{V}_{ij}$ becomes a straight line, i.e., the radius of the corresponding Apollonius circle r_{ij}^{AC} becomes infinite.

Proof. See [19]. $\square$

In order to obtain the decentralized dynamics, we analyze the influence of each individual predator's location on the cost function (15). Based on the distance metric (17), the cost function (15) can be rewritten as

$$\begin{aligned} F(\mathbf{p}, t) = & \sum_{n \in \mathcal{M}_{\mathcal{N}}^i} \int_{\mathcal{V}_{i,n}(\mathbf{p})} w_{i,n}^2 \|\mathbf{p}_i - \mathbf{q}\|^2 \phi_n(\mathbf{q}, t) d\mathbf{q} \\ & + \sum_{n \in \mathcal{M}_{\mathcal{N}}^i} \sum_{j \in \mathcal{N}_{\mathcal{V}_{i,n}}} \int_{\mathcal{V}_{j,n}(\mathbf{p})} w_{j,n}^2 \|\mathbf{p}_j - \mathbf{q}\|^2 \phi_n(\mathbf{q}, t) d\mathbf{q} \\ & + \sum_{n \in \mathcal{M}_{\mathcal{N}}^i} \sum_{\substack{j \in \mathcal{N}_{\mathcal{M}}^n \\ j \notin \mathcal{N}_{\mathcal{V}_{i,n}}} } \int_{\mathcal{V}_{j,n}(\mathbf{p})} w_{j,n}^2 \|\mathbf{p}_j - \mathbf{q}\|^2 \phi_n(\mathbf{q}, t) d\mathbf{q}, \quad (20) \end{aligned}$$

in which $\mathcal{N}_{\mathcal{V}_{i,n}}$ and $\bar{\mathcal{N}}_{\mathcal{V}_{i,n}} := \mathcal{N}_{\mathcal{V}_{i,n}} \cup \{i\}$ stand for the index sets of the Delaunay neighbors and closed Delaunay neighbors of Predator i with respect to Prey Species n in the multiplicatively weighted Voronoi diagram, respectively.

The first term in (20) encodes the cost for Predator i , and the second term encodes the cost for the predators sharing Voronoi boundaries (given by Lemma V.1) with Predator i , i.e., the 1-hop neighbors of Predator i in the Voronoi diagram. Since the last term in (20) encodes the cost for the predators that do not share Voronoi boundaries with Predator i , it does not depend on the location of Predator i . Hence, only the first two terms in (20) remain in the partial derivative of $F(\mathbf{p}, t)$ with respect to $\mathbf{p}_i$, resulting in

$$\begin{aligned} & \frac{\partial}{\partial \mathbf{p}_i} F(\mathbf{p}, t) \\ = & \frac{\partial}{\partial \mathbf{p}_i} \sum_{n \in \mathcal{M}_{\mathcal{N}}^i} \int_{\mathcal{V}_{i,n}(\mathbf{p})} w_{i,n}^2 \|\mathbf{p}_i - \mathbf{q}\|^2 \phi_n(\mathbf{q}, t) d\mathbf{q} \\ & + \frac{\partial}{\partial \mathbf{p}_i} \sum_{n \in \mathcal{M}_{\mathcal{N}}^i} \sum_{j \in \mathcal{N}_{\mathcal{V}_{i,n}}} \int_{\mathcal{V}_{j,n}(\mathbf{p})} w_{j,n}^2 \|\mathbf{p}_j - \mathbf{q}\|^2 \phi_n(\mathbf{q}, t) d\mathbf{q}. \quad (21) \end{aligned}$$

Applying Lemma IV.1 to (21) results in

$$\frac{\partial}{\partial \mathbf{p}_i} F(\mathbf{p}, t) = \sum_{n \in \mathcal{M}_{\mathcal{N}}^i} (2w_{i,n}^2 m_{i,n}(\mathbf{p}, t) \cdot (\mathbf{p}_i - \mathbf{c}_{i,n}(\mathbf{p}, t))^\top), \quad (22)$$

where $m_{i,n}(\mathbf{p}, t)$ and $\mathbf{c}_{i,n}(\mathbf{p}, t)$ are defined as

$$m_{i,n}(\mathbf{p}, t) := \int_{\mathcal{V}_{i,n}(\mathbf{p})} \phi_n(\mathbf{q}, t) d\mathbf{q} \quad (23)$$

and

$$\mathbf{c}_{i,n}(\mathbf{p}, t) := \frac{\int_{\mathcal{V}_{i,n}(\mathbf{p})} \mathbf{q} \phi_n(\mathbf{q}, t) d\mathbf{q}}{m_{i,n}(\mathbf{p}, t)} \quad (24)$$

respectively. According to (22), a critical point of $F(\mathbf{p}, t)$ with respect to $\mathbf{p}_i$ can be calculated as the weighted center of mass of Predator i 's dominant subdomains,

$$\mathbf{C}_i^w(\mathbf{p}, t) = \frac{\sum_{n \in \mathcal{M}_{\mathcal{N}}^i} (w_{i,n}^2 m_{i,n}(\mathbf{p}, t) \mathbf{c}_{i,n}(\mathbf{p}, t))}{\sum_{n \in \mathcal{M}_{\mathcal{N}}^i} (w_{i,n}^2 m_{i,n}(\mathbf{p}, t))}. \quad (25)$$

With dynamic and changing environmental conditions, the controller that can adaptively stabilize the system to the time-varying goal configuration (25) for all $i \in \mathcal{N}$ should be derived. Hence, we analyze the Lyapunov function

$$V_i(\mathbf{p}, t) = \frac{1}{2} \|\mathbf{p}_i - \mathbf{C}_i^w(\mathbf{p}, t)\|^2. \quad (26)$$

Note that the seemingly innocent-looking (26) does not imply that a simple controller pointing each predator towards $\mathbf{C}_i^w(\mathbf{p}, t)$ results in $\lim_{t \rightarrow \infty} V_i(\mathbf{p}, t) = 0$ because $\frac{d}{dt} V_i(\mathbf{p}, t)$ involves keeping track of the rate of change of the boundaries of all the corresponding Voronoi cells.

To find the control law such that $\lim_{t \rightarrow \infty} V_i(\mathbf{p}, t) = 0$, $\forall i \in \mathcal{N}$, we adopt the concept of liveness, e.g., [13], and adapt it to the time-varying setting through the notion of time-varying control Lyapunov function (TV-CLF) defined below (see also [44] on CLFs for time-varying systems).

Definition V.1. $V(\mathbf{x}, t) \in C^1 : \mathcal{X} \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is a TV-CLF with respect to the dynamical system (1) if there exists a locally Lipschitz class $\mathcal{K}$ function $\alpha : [0, a) \rightarrow \mathbb{R}_{\geq 0}$, in which $a < \infty$, such that

$$\inf_{\mathbf{u} \in \mathcal{U}} \left[\frac{d}{dt} V(\mathbf{x}, t) \right] \leq -\alpha(V(\mathbf{x}, t)),$$

for all $\mathbf{x} \in \mathcal{X}$, where

$$\frac{d}{dt} V(\mathbf{x}, t) = \mathcal{L}_f V(\mathbf{x}, t) + \mathcal{L}_G V(\mathbf{x}, t) \mathbf{u} + \frac{\partial}{\partial t} V(\mathbf{x}, t),$$

in which $\mathcal{L}$ denotes the Lie derivative.

Then, we have the following lemma that is essential to construct the heterogeneous predator dynamics.

Lemma V.2. If there exists a TV-CLF, $V(\mathbf{x}, t)$, as per Definition V.1, then the system (1) will be asymptotically stabilized to $\mathbf{x}^*(t)$ such that $V(\mathbf{x}^*(t), t) = 0$ by any locally Lipschitz continuous $\mathbf{u} \in \mathcal{U}_{\text{CLF}}$, where

$$\mathcal{U}_{\text{CLF}} = \left\{ \mathbf{u} \in \mathcal{U} \mid \frac{d}{dt} V(\mathbf{x}, t) \leq -\alpha(V(\mathbf{x}, t)) \right\}.$$

Proof. (Paralleling the proof machinery in [45]) Consider a reference function $\bar{V}$ satisfying the scalar differential equation

$$\dot{\bar{V}} + \alpha(\bar{V}) = 0, \text{ with } \bar{V}(t_0) = V(\mathbf{x}(t_0), t_0), \quad (27)$$

in which $\alpha : [0, a) \rightarrow \mathbb{R}_{\geq 0}$, where $a < \infty$, is a locally Lipschitz class $\mathcal{K}$ function. By Lemma 4.4 of [46], for all $0 < V(\mathbf{x}(t_0), t_0) < a$, (27) has a unique solution $\bar{V}(t)$ for all $t \geq t_0$ (assuming forward completeness) and

$$\bar{V}(t) = \bar{\alpha}(V(\mathbf{x}(t_0), t_0), t - t_0), \quad (28)$$

where $\bar{\alpha} : [0, a) \times [0, \infty) \rightarrow [0, \infty)$ is a class $\mathcal{KL}$ function. According to (27) and (28), by Lemma 3.4 of [46], we obtain

$$V(\mathbf{x}(t), t) \leq \bar{\alpha}(V(\mathbf{x}(t_0), t_0), t - t_0)$$

for all $t \geq t_0$. Since $\lim_{t \rightarrow \infty} \bar{\alpha}(V(\mathbf{x}(t_0), t_0), t - t_0) = 0$, then $\lim_{t \rightarrow \infty} V(\mathbf{x}(t), t) = 0$, i.e., $\lim_{t \rightarrow \infty} \mathbf{x}(t) = \mathbf{x}^*(t)$. $\square$

In order for the team of predators to asymptotically achieve a time-varying local minimum of (15) with respect to $\mathbf{p}$, similar to [12] that synthesizes the controller through a quadratic

program, we synthesize the TV-CLF-based controller for the team of predators in Theorem V.1.

Theorem V.1. (Heterogeneous Predator Dynamics) For a team of robots each of which satisfies $\dot{\mathbf{p}}_i = \mathbf{f}(\mathbf{p}_i) + G(\mathbf{p}_i) \mathbf{u}_i$, if each robot is executing the controller $\mathbf{u}_i$ resulting from the TV-CLF-based quadratic program

$$\begin{aligned} \arg \min_{\mathbf{u}_i \in \mathbb{R}^q} \quad & \|\mathbf{u}_i\|^2 \\ \text{s.t.} \quad & \frac{d}{dt} V_i(\mathbf{p}, t) \leq -\alpha(V_i(\mathbf{p}, t)) \end{aligned} \quad (29)$$

where $\alpha : [0, a) \rightarrow \mathbb{R}_{\geq 0}$, with $a < \infty$, is a subadditive class $\mathcal{K}$ function,

$$\frac{d}{dt} V_i(\mathbf{p}, t) = \mathcal{L}_{f_i} V_i(\mathbf{p}, t) + \mathcal{L}_G V_i(\mathbf{p}, t) \mathbf{u}_i + \frac{\partial}{\partial t} V_i(\mathbf{p}, t)$$

in which f_i consists of $\mathbf{f}(\mathbf{p}_i)$ and $\dot{\mathbf{p}}_j$, $\forall j \in \cup_{n \in \mathcal{M}_N^i} \mathcal{N}_{V_{i,n}}$,

$$\begin{aligned} \frac{\partial}{\partial \mathbf{p}_i} V_i(\mathbf{p}, t) &= (\mathbf{p}_i - \mathbf{C}_i^w(\mathbf{p}, t))^\top \\ &\cdot \left(I_d - \frac{1}{\sum_{n \in \mathcal{M}_N^i} (w_{i,n}^2 m_{i,n}(\mathbf{p}, t))} \sum_{n \in \mathcal{M}_N^i} \left(w_{i,n}^2 \sum_{k \in \mathcal{N}_{V_{i,n}}} \right. \right. \\ &\left. \left. \left(\int_{\partial V_{ik,n}(\mathbf{p})} \frac{w_{i,n}^2 (\mathbf{q} - \mathbf{C}_i^w(\mathbf{p}, t)) (\mathbf{q} - \mathbf{p}_i)^\top}{\|w_{i,n}^2 (\mathbf{q} - \mathbf{p}_i) - w_{k,n}^2 (\mathbf{q} - \mathbf{p}_k)\|} \phi_n(\mathbf{q}, t) d\mathbf{q} \right) \right) \right) \end{aligned} \quad (30)$$

in which $\partial V_{ik,n}(\mathbf{p})$ stands for the bisector, determined by Lemma V.1, between Predators i and k with respect to Prey Species n in the multiplicatively weighted Voronoi diagram, and

$$\begin{aligned} \frac{\partial}{\partial t} V_i(\mathbf{p}, t) &= \frac{(\mathbf{C}_i^w(\mathbf{p}, t) - \mathbf{p}_i)^\top}{\sum_{n \in \mathcal{M}_N^i} (w_{i,n}^2 m_{i,n}(\mathbf{p}, t))} \\ &\cdot \sum_{n \in \mathcal{M}_N^i} \left(w_{i,n}^2 \int_{V_{i,n}(\mathbf{p})} (\mathbf{q} - \mathbf{C}_i^w(\mathbf{p}, t)) \frac{\partial}{\partial t} \phi_n(\mathbf{q}, t) d\mathbf{q} \right) \end{aligned} \quad (31)$$

in which $\frac{\partial}{\partial t} \phi_n(\mathbf{q}, t)$ is determined by the heterogeneous prey dynamics (9), then the team of agents asymptotically achieves $\mathbf{C}_i^w(\mathbf{p}, t)$ defined by (25) for all $i \in \mathcal{N}$, i.e., a local minimum of (15) with respect to $\mathbf{p}$.

Proof. Denote the TV-CLF for the entire team of agents as

$$V(\mathbf{p}, t) = \frac{1}{2} \sum_{i \in \mathcal{N}} \|\mathbf{p}_i - \mathbf{C}_i^w(\mathbf{p}, t)\|^2 = \sum_{i \in \mathcal{N}} V_i(\mathbf{p}, t),$$

whose total time derivative is

$$\frac{d}{dt} V(\mathbf{p}, t) = \sum_{i \in \mathcal{N}} \frac{d}{dt} V_i(\mathbf{p}, t).$$

According to (29), $\frac{d}{dt} V_i(\mathbf{p}, t) \leq -\alpha(V_i(\mathbf{p}, t))$, and since $\alpha(\cdot)$ is assumed to be a subadditive class $\mathcal{K}$ function, then we have

$$\frac{d}{dt} V(\mathbf{p}, t) \leq - \sum_{i \in \mathcal{N}} \alpha(V_i(\mathbf{p}, t)) \leq -\alpha(V(\mathbf{p}, t)).$$

Based on Lemma V.2, we have

$$\lim_{t \rightarrow \infty} V(\mathbf{p}, t) = 0,$$

so, according to (22) and (25), the entire team of agents asymptotically achieves $\mathbf{p}^*(t)$ such that $\frac{\partial F}{\partial \mathbf{p}}(\mathbf{p}^*(t), t) = \mathbf{0}^\top$.

Moreover, according to (16), (17), (23), (24), (25), and (26), utilizing Lemma IV.1 and Lemma V.1, (30) and (31) can be obtained. $\square$

Note that although the arguments of the quantities $V_i(\mathbf{p}, t)$, $C_i^w(\mathbf{p}, t)$, $m_{i,n}(\mathbf{p}, t)$, $\partial \mathcal{V}_{ik,n}(\mathbf{p})$, and $\mathcal{V}_{i,n}(\mathbf{p})$ in Theorem V.1 contain $\mathbf{p}$ for notational simplicity, they actually depend only on the closed neighborhood of $\mathbf{p}_i$ in the Delaunay graph of the multiplicatively-weighted Voronoi diagram with respect to each $\phi_n(\mathbf{q}, t)$. Moreover, for each agent to obtain its controller $\mathbf{u}_i$, in addition to the states of its neighboring agents, it needs the information of its target prey and the dynamics of its target prey within its corresponding Voronoi cell(s), as seen in (30) and (31) where (9) comes into play. In summary, the information to which each agent needs access is the states of its Delaunay neighbors and its target prey density (or densities) and prey dynamics within its corresponding Voronoi cell(s), as seen in (16), (23), (24), (25), (30) and (31). As such, the heterogeneous predator dynamics (29) is decentralized and achieves Assumption (b).

Remark V.1. Only when the out-degree of each predator vertex in $\mathcal{G}_{\text{prey}}^{\text{pred}}$ defined for an application is 1, the controllers obtained from Theorem V.1 will drive the predators pursuing Prey Species n to a time-varying centroidal Voronoi tessellation with respect to $\phi_n(\mathbf{q}, t)$, which achieves similar results as [17], [26]. Otherwise, the resulting time-varying optimal configuration is not a centroidal Voronoi tessellation in general as what can be observed in (25).

According to Theorem V.1, the predators dynamically move to the areas with high density of their target prey species. This implies that if a predator pursues one prey species, it asymptotically achieves the time-varying center of mass of its dominant subdomain with respect to its target prey. If a predator pursues multiple prey species, it asymptotically achieves a time-varying optimal position that balances every species of its target prey, i.e., a weighted center of mass of its dominant subdomains with respect to all species of its target prey, as defined by (25). Therefore, Assumption (c) is achieved. So far, all the assumptions presented at the beginning of this section have been achieved by the heterogeneous predator dynamics (29) given in Theorem V.1.

When a team of predators are executing the dynamics resulting from Theorem V.1, their safety should also be considered. Inspired by [13], we utilize control barrier functions (CBFs) to encode safety as set forward invariance.

Table II
ADDITIONAL HETEROGENEOUS MODALITIES IN (32)

Symbol	Description
$\mathcal{G}_{\text{prey}}^{\text{pred}}$	Encoding of task suitability among different agents
$w_{i,n}$	Quantification of heterogeneous relative performance
$h_{i,s}$	Characterization of safe operating regions

Definition V.2. (Definition 4 in [47] following [48], [49]) $h(\mathbf{x}, t) \in C^1 : \mathcal{X} \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ is a time-varying control barrier function (TV-CBF) with respect to the dynamical system (1) if there exists a locally Lipschitz extended class $\mathcal{K}_\infty$ function $\gamma : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\sup_{\mathbf{u} \in \mathcal{U}} \left[\frac{d}{dt} h(\mathbf{x}, t) \right] \geq -\gamma(h(\mathbf{x}, t)),$$

for all $\mathbf{x} \in \mathcal{X}$, where

$$\frac{d}{dt} h(\mathbf{x}, t) = \mathcal{L}_f h(\mathbf{x}, t) + \mathcal{L}_G h(\mathbf{x}, t) \mathbf{u} + \frac{\partial}{\partial t} h(\mathbf{x}, t).$$

Lemma V.3. If there exists a TV-CBF, $h(\mathbf{x}, t)$, as per Definition V.2, and 0 is a regular value of $h(\mathbf{x}, t)$, then the set, $\mathcal{S}(t) = \{\mathbf{x} \in \mathcal{X} \mid h(\mathbf{x}, t) \geq 0\}$, is rendered forward invariant by any locally Lipschitz continuous $\mathbf{u} \in \mathcal{U}$ such that $\mathcal{L}_f h(\mathbf{x}) + \mathcal{L}_G h(\mathbf{x}) \mathbf{u} + \frac{\partial}{\partial t} h(\mathbf{x}, t) \geq -\gamma(h(\mathbf{x}))$.

Proof. See [47]. $\square$

The priority over liveness (task achievement) encoded by the TV-CLF, is safety encoded by TV-CBFs. However, since liveness can sometimes conflict with safety, similar to [13], we relax the liveness constraint by introducing a relaxation variable to it. Combining the heterogeneous prey and predator dynamics based on Proposition IV.1 and Theorem V.1, we propose the following collaborative pursuit framework through adapting the CLF-CBF-based quadratic program discussed in [13] to the time-varying setting.

Proposition V.1. The ecologically inspired collaborative pursuit framework is formulated as the TV-CLF-CBF-based quadratic program, where each robot aims to minimize its control effort subject to liveness and safety constraints,

$$\begin{aligned} & \arg \min_{(\mathbf{u}_i, \delta_i) \in \mathbb{R}^{q+1}} \|\mathbf{u}_i\|^2 + \kappa_i \delta_i^2 \\ & \text{s.t.} \quad \mathcal{L}_f V_i + \mathcal{L}_G V_i \mathbf{u}_i + \frac{\partial}{\partial t} V_i \leq -\alpha(V_i) + \delta_i \\ & \quad \mathcal{L}_f h_{i,s} + \mathcal{L}_G h_{i,s} \mathbf{u}_i + \frac{\partial}{\partial t} h_{i,s} \geq -\gamma(h_{i,s}) \end{aligned} \quad (32)$$

$\forall s \in \{1, 2, \dots, S\}$ with $S \in \mathbb{Z}^+$, in which the arguments are omitted for concision of the expression, the s^{th} CBF of Robot i , $h_{i,s}(\mathbf{p}_i, t)$, should be determined based on specific application requirements, and V_i is given by (26).

In addition to the eight heterogeneous modalities listed in Table I, three more listed in Table II are also encapsulated in (32). Leveraging the proposed heterogeneous multi-robot collaborative pursuit framework given by Proposition V.1, various tasks can be accomplished with certain parameter specifications.

VI. APPLICATIONS

In this section, we conduct experiments on four applications² of the proposed collaborative pursuit framework (32) by specifying the liveness constraint and the safety constraints.

²A video containing the four full experiments is provided in the supplementary multimedia files.

These applications range from multi-agent coordination to collaboration, interpreted in a slightly different perspective from [50]. Note that the main benefit of providing these applications is twofold. First, the adaptivity of the proposed framework that can be applied to a wide range of tasks, i.e., adaptivity (B) mentioned in Section I, is demonstrated. Second, these applications with different parameter specifications of (32) themselves serve as comparison and ablation experiments for each other.

The common threads through the four applications are as follows. The experiments are carried out using multiple differential-drive wheeled robots in the Robotarium [51], where the domain of interest $\mathcal{D} \subset \mathbb{R}^2$ is a rectangular domain with the x -axis ranging from $B^l = -1.6$ meters to $B^r = 1.6$ meters and the y -axis ranging from $B^d = -1$ meter to $B^u = 1$ meter. We assume single integrator dynamics for $\mathbf{p}_i$, i.e., $\mathbf{f}(\mathbf{p}_i) = \mathbf{0}_2$, $G(\mathbf{p}_i) = I_2$, $\forall i \in \mathcal{N}$, corresponding to a point near the axle of the robot [52]. In Table I, the units of the parameters that are only related to space scale, e.g., $\chi_{n,i}$ and $r_{n,i}^R$, are in meters. The parameters related to time scale enter into dimensionless combinations with spatial and temporal resolutions, so they do not represent the physical time scale. Both zero Neumann and Dirichlet boundary conditions are assumed on $\partial\mathcal{D}$. Since each robot should move within $\mathcal{D}$ in a collision-free manner, we set five CBFs for every application,

$$h_{i,1}(\mathbf{p}_i) = [1, 0] \cdot \mathbf{p}_i - B^l, \quad (33)$$

$$h_{i,2}(\mathbf{p}_i) = B^r - [1, 0] \cdot \mathbf{p}_i, \quad (34)$$

$$h_{i,3}(\mathbf{p}_i) = [0, 1] \cdot \mathbf{p}_i - B^d, \quad (35)$$

$$h_{i,4}(\mathbf{p}_i) = B^u - [0, 1] \cdot \mathbf{p}_i, \quad (36)$$

$$h_{i,5}(\mathbf{p}_i, t) = \|\mathbf{p}_i - \mathbf{p}_j(t)\|^2 - D_{ij}^2, \quad \forall j \neq i \in \mathcal{N}_{\text{CA}}^i, \quad (37)$$

$\forall i \in \mathcal{N}$, where (33), (34), (35), and (36) are to constrain robots within $\mathcal{D}$, and (37) encodes the collision avoidance region, in which $D_{ij} \in \mathbb{R}_{>0}$ is the safe distance between Robots i and j , and $\mathcal{N}_{\text{CA}}^i \subset \mathcal{N}$ is the index set of the neighbors of Robot i , with which Robot i must avoid collisions.

A. Forest Firefighting

In this subsection, we apply the framework (32) to a forest firefighting task, considered as a case of multi-agent coordination, where a team of agents organizes its actions to perform a task coherently.

Specifically, we consider $M = 1$ type (species) of forest fire (prey) to be suppressed (pursued) by a team of $N = 10$ robots, where the prey density $\phi_1(\mathbf{q}, t)$ is thermal energy. The corresponding predator-prey graph is shown in Fig. 6. Based on [53], we assume the intrinsic reaction process $R_1(\mathbf{q}, t)$ in (9) to be governed by the loss of thermal energy to the environment and combustion of the vegetation, i.e.,

$$R_1(\mathbf{q}, t) = k^{\text{loss}} \cdot (T^{\text{env}} - \phi_1(\mathbf{q}, t)) + k^{\text{comb}} \cdot f^{\text{comb}}(\mathbf{q}, t),$$

where $k^{\text{loss}} \in \mathbb{R}_{\geq 0}$ encodes how fast the thermal energy is lost to the environment, $T^{\text{env}} \in \mathbb{R}_{\geq 0}$ denotes the thermal energy of the environment, which can be considered as the thermal energy reference point, and $k^{\text{comb}} \in \mathbb{R}_{\geq 0}$ denotes the

combustion coefficient encoding how fast the thermal energy is generated by the process of fuel combustion. We assume the combustion fuel to be the vegetation whose dynamics is

$$\frac{\partial}{\partial t} \phi^v(\mathbf{q}, t) = -f^{\text{comb}}(\mathbf{q}, t),$$

where the combustion function is defined as

$$f^{\text{comb}}(\mathbf{q}, t) = \begin{cases} 0 & \text{if Cond}_1 \\ k^{\text{burn}} \cdot \phi^v(\mathbf{q}, \tau_2) & \text{if Cond}_2 \\ k^{\text{burn}} \cdot \phi^v(\mathbf{q}, t) & \text{if Cond}_3 \end{cases} \quad (38)$$

in which the conditions are specifically defined as: $\text{Cond}_1 = (\phi_1(\mathbf{q}, \tau_1) < T^{\text{ign}}, \forall \tau_1 \leq t) \vee (\exists \tau_1 \leq t, \text{s.t. } \phi^v(\mathbf{q}, \tau_1) = 0)$, where $T^{\text{ign}} \in \mathbb{R}_{\geq 0}$ denotes the reference ignition thermal energy and $\tau_1 \in \mathbb{R}_{\geq 0}$ is a temporal variable, $\text{Cond}_2 = (\phi_1(\mathbf{q}, t) < T^{\text{ign}})$ with τ_2 being the solution to $\arg \min_{\tau_2} \|\tau_2 - t\|$ subject to $(\tau_2 < t) \wedge (\phi_1(\mathbf{q}, \tau_2) = T^{\text{ign}})$, and $\text{Cond}_3 = (\phi_1(\mathbf{q}, t) \geq T^{\text{ign}})$. Note that as forest fire modeling is complex in general and still an active area of research, considering miscellaneous physical processes and influencing factors, there are many alternative methods to model fire propagation, e.g., [54]–[56].

Since the diffusion process can be affected by environmental factors such as topography and moisture, we let the diffusivity D_{11} be space-dependent and defined as $D_{11}(\mathbf{q}) = [5 \times 10^{-6}, 0] \cdot \mathbf{q} + 4 \times 10^{-5}$. Additionally, since the firefighting robots (predators) do not directly influence the advection (drift) process of the thermal energy (prey), we let $\Lambda_{1,i} = 0_2$, $\forall i \in \mathcal{N}_{\mathcal{M}}^1$. Moreover, since there can be wind that governs the advection process of $\phi_1(\mathbf{q}, t)$, we let the vector field $\Gamma_1 = [10^{-3}, 0]^\top$. Without loss of generality, we let $T^{\text{env}} = 0$ as a reference point. In addition, we let $k^{\text{loss}} = 2 \times 10^{-2}$, $k^{\text{comb}} = 20$, $k^{\text{burn}} = 1 \times 10^{-2}$. Regarding the influence of the firefighting robots on the reaction process of the forest fire, we let $f_{1,i}^R = -\sigma_{1,i} \cdot \phi_1(\mathbf{q}, t)$, $\forall i \in \mathcal{N}$, where $\sigma_{1,1} = 4 \times 10^{-2}$, $\sigma_{1,2} = 4 \times 10^{-2}$, $\sigma_{1,3} = 3 \times 10^{-2}$, $\sigma_{1,4} = 3 \times 10^{-2}$, $\sigma_{1,5} = 2 \times 10^{-2}$, $\sigma_{1,6} = 2 \times 10^{-2}$, $\sigma_{1,7} = 3 \times 10^{-2}$, $\sigma_{1,8} = 3 \times 10^{-2}$, $\sigma_{1,9} = 4 \times 10^{-2}$, $\sigma_{1,10} = 4 \times 10^{-2}$. In addition, we let $\delta_{1,i}^R = 10^{-3}$, $\forall i \in \mathcal{N}$, and $r_{1,1}^R = 0.08$, $r_{1,2}^R = 0.1$, $r_{1,3}^R = 0.12$, $r_{1,4}^R = r_{1,5}^R = r_{1,6}^R = r_{1,7}^R = r_{1,8}^R = r_{1,9}^R = r_{1,10}^R = 0.14$. Regarding the weighting metrics encoding the relative performance among the firefighting robots with respect to ϕ_1 , we let $w_{1,1} = 1/6$, $w_{2,1} = 1/7$, $w_{3,1} = 1/8$, $w_{4,1} = 1/9$, $w_{5,1} = 1/9$, $w_{6,1} = 1/8$, $w_{7,1} = 1/7$, $w_{8,1} = 1/6$, $w_{9,1} = 1/5$, and $w_{10,1} = 1/5$. The initial positions of the robots are shown in Fig. 7(a), and we let $\phi^v(\mathbf{q}, 0) = 50$, and $\phi_1(\mathbf{q}, 0) = 10^3 \cdot \exp\left(-\frac{\|\mathbf{q} - [-0.5, 0]^\top\|^2}{2 \times 0.15^2}\right)$.

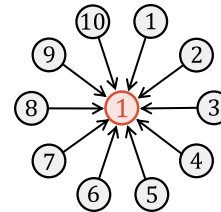


Figure 6. The predator-prey graph for the forest firefighting application in Section VI-A.

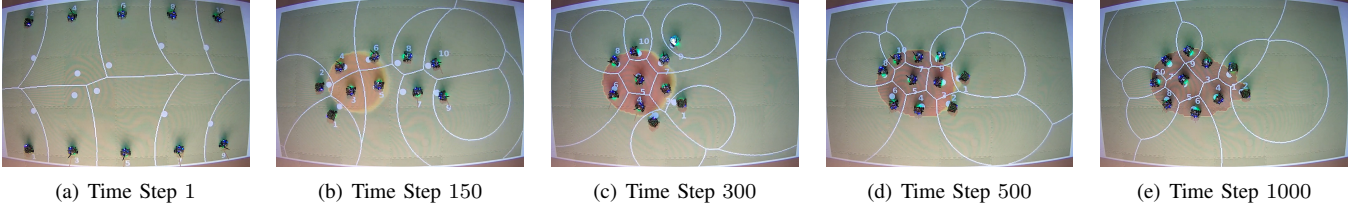


Figure 7. Snapshots of the deployment of ten robots to put out simulated forest fire projected onto the testbed, with the green area representing higher density of the vegetation $\phi^v(\mathbf{q}, t)$ and the dark red area representing lower density of the vegetation $\phi^v(\mathbf{q}, t)$. Each robot is in charge of a dominant subdomain (Voronoi cell) with a set of white curves as its boundaries and a white dot as its center of mass with respect to the thermal energy $\phi_1(\mathbf{q}, t)$.

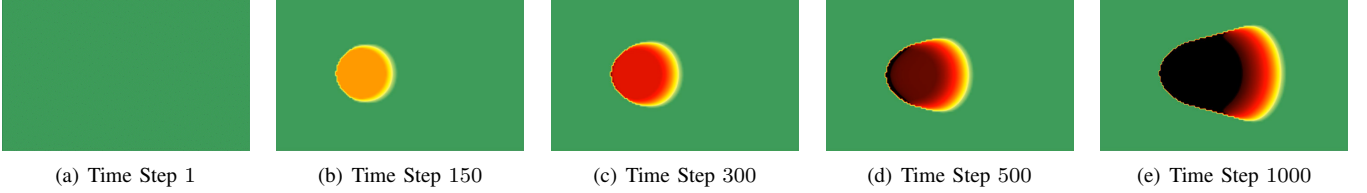


Figure 8. The development of the vegetation without the influence of firefighting robots, where the green area indicates higher vegetation density and the dark red area indicates lower vegetation density. The forest fire is gradually consuming the vegetation.

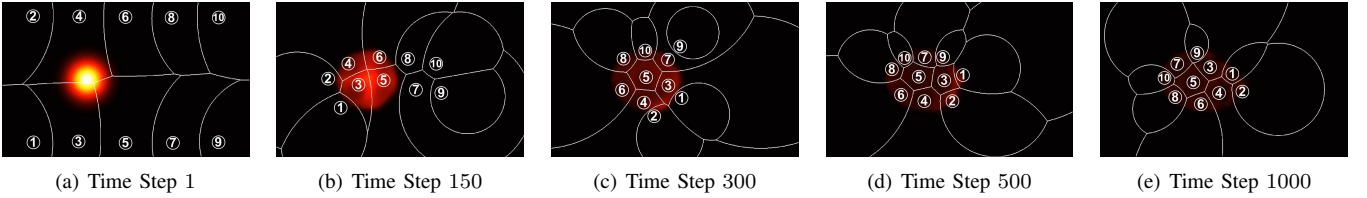


Figure 9. The development of the thermal energy $\phi_1(\mathbf{q}, t)$ corresponding to Fig. 7, with the bright yellow area representing higher thermal energy and the dark red area representing lower thermal energy. The robots are indexed by corresponding numbers and the boundaries of their dominant subdomains are represented by the white curves that are the same as Fig. 7.

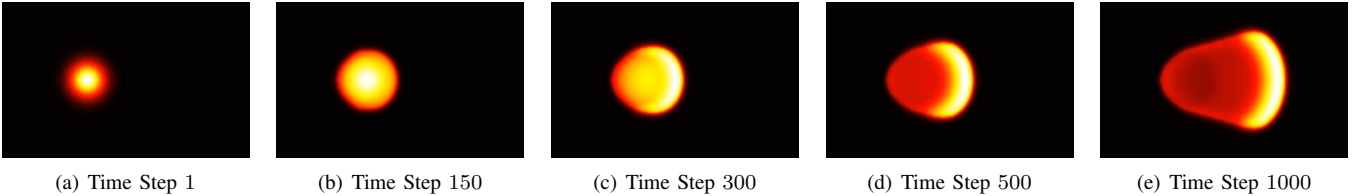


Figure 10. The development of the thermal energy without the influence of firefighting robots, where the bright yellow area indicates higher thermal energy and the dark red area indicates lower thermal energy. The thermal energy is progressively spreading.

The experimental results are shown in Fig. 7, Fig. 9, Fig. 8, Fig. 10, and Fig. 11. The evolution of the thermal energy $\phi_1(\mathbf{q}, t)$ under the influence of firefighting robots is shown in Fig. 9, where the firefighting robots are asymptotically optimally covering $\phi_1(\mathbf{q}, t)$, as shown in Fig. 11(a), which verifies Theorem V.1. In addition, for comparison purposes, the corresponding evolution of $\phi^v(\mathbf{q}, t)$ and $\phi_1(\mathbf{q}, t)$ without the influence of firefighting robots are shown in Fig. 8 and Fig. 10, respectively. Compared with Fig. 7 and Fig. 9, one can observe that the firefighting robots have suppressed the development of the forest fire that would otherwise intrinsically burn down the vegetation in the forest, which can also be seen in Fig. 11(b), representing the evolution of the entire thermal energy in $\mathcal{D}$, and Fig. 11(c), representing the evolution of the total amount of vegetation in $\mathcal{D}$.

In addition, as shown in Fig. 11(b), in the case without firefighting robots, the total thermal energy first increases from time step 1 to around step 60 due to the combustion process of the vegetation generating additional thermal energy, and then decreases from around time step 60 to step 400 due to the relatively stronger effect of thermal energy loss to the environment than that of combustion of the vegetation. Afterwards, the thermal energy continues to increase, resulting in the spread of the forest fire, as also shown in Fig. 10. Moreover, in the case with firefighting robots, the total thermal energy also increases from time step 1 to around step 60 because the firefighting robots are not close to an optimal coverage over the thermal energy so the “consuming” effect has not been fully realized, while from around time step 60 onward, as the robots asymptotically achieve a time-varying

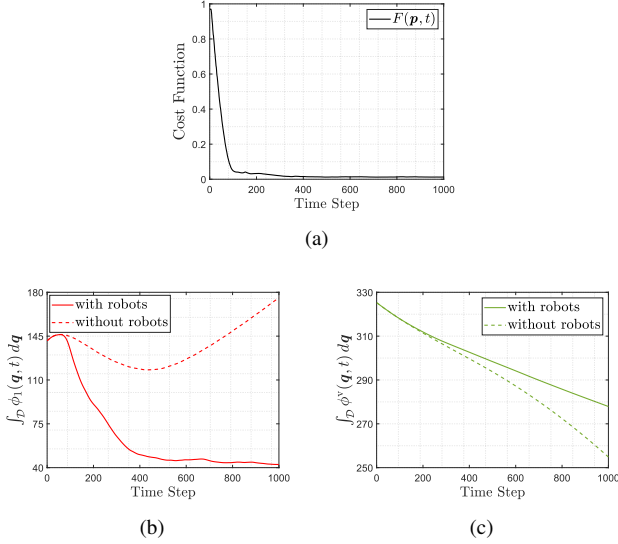


Figure 11. Experimental results of the forest firefighting application in Section VI-A. (a) Evolution of the cost $F(\mathbf{p}, t)$; (b) Comparison of the evolution of the entire thermal energy over $\mathcal{D}$ between the cases with and without firefighting robots; (c) Comparison of the evolution of the total amount of vegetation over $\mathcal{D}$ between the cases with and without firefighting robots.

optimal coverage configuration, as seen in Fig. 7, Fig. 9, and Fig. 11(a), the total thermal energy across $\mathcal{D}$ asymptotically decreases, which verifies Proposition IV.2.

B. Pursuit-Evasion

In the application of (32) to forest firefighting in Section VI-A, the predators do not influence the drift process of the prey because the firefighting robots do not influence the advection process of the forest fire. Additionally, all the predators have the same task suitability.

In this subsection, we demonstrate how the predators influence the drift process of the prey, and how they leverage their different task suitability through a novel pursuit-evasion scheme resulting from (32). This scheme is considered as multi-agent cooperation, where agents operate in their respective roles, willingly accommodating each other.

First, we apply the pursuit-evasion scheme to the case with one predator and one prey species to particularly demonstrate how the predator's state affects the drift process of the prey. Then, we apply it to a more complex situation with multiple predators pursuing two prey species.

In the first experiment with one predator whose position is denoted as $\mathbf{p}_0(t)$ pursuing one prey species whose density distribution is denoted as $\phi_0(\mathbf{q}, t)$, we assume no reaction process and the intrinsic motion of the prey to be governed by an augmented drift-diffusion equation

$$\frac{\partial}{\partial t} \phi_0 + \langle \nabla_{\mathbf{q}}, \tilde{\Gamma}_0^T \phi_0 - D_{00} \nabla_{\mathbf{q}} \phi_0 \rangle = 0,$$

where $D_{00} = 1$, $\phi_0(\mathbf{q}, 0) = \exp\left(-\frac{\|\mathbf{q} - [-0.4, 0.2]^T\|^2}{2 \times 0.1^2}\right)$, and $\Gamma_0(\mathbf{q}) = \begin{bmatrix} 0 & -70 \\ 30 & 0 \end{bmatrix} \cdot \mathbf{q}$. With the influence of the predator, the drift process of $\phi_0(\mathbf{q}, t)$ is augmented to $\tilde{\Gamma}_0(\mathbf{q}, \mathbf{p}_0, \dot{\mathbf{p}}_0) = \Gamma_0(\mathbf{q}) + \exp\left(-\frac{\|\mathbf{p}_0 - \mathbf{q}\|^2}{\chi_{0,0}^2}\right) \Lambda_{0,0} \dot{\mathbf{p}}_0(t)$, in

which $\Lambda_{0,0} = \text{diag}(700, 300)$ and $\chi_{0,0} = 0.4$. The experimental results are shown in Fig. 13, demonstrating how the predator pursues the prey (without consuming the prey) and how the prey evade the predator.

In the second experiment including reaction processes, we consider $M = 2$ species (types) of prey (evaders) denoted as $\phi_1(\mathbf{q}, t)$ and $\phi_2(\mathbf{q}, t)$ to be pursued by a team of $N = 7$ heterogeneous predators (robots), where Predators 1, 2, 3, and 4 pursue $\phi_1(\mathbf{q}, t)$, while Predators 5, 6, and 7 pursue $\phi_2(\mathbf{q}, t)$. The corresponding predator-prey graph is shown in Fig. 12. The prey dynamics (9) specifically becomes

$$\frac{\partial}{\partial t} \begin{bmatrix} \phi_1 \\ \phi_2 \end{bmatrix} + \nabla_{\mathbf{q}} \circ \left(\begin{bmatrix} \tilde{\Gamma}_1^T \phi_1 \\ \tilde{\Gamma}_2^T \phi_2 \end{bmatrix} - \begin{bmatrix} D_{11} & D_{12} \\ D_{21} & D_{22} \end{bmatrix} \nabla_{\mathbf{q}} \begin{bmatrix} \phi_1 \\ \phi_2 \end{bmatrix} \right) = \begin{bmatrix} \tilde{R}_1 \\ \tilde{R}_2 \end{bmatrix}$$

in which: (i) Regarding the augmented drift terms,

$$\tilde{\Gamma}_1(\mathbf{q}, \mathbf{p}, \dot{\mathbf{p}}) = \Gamma_1(\mathbf{q}) + \sum_{i=1}^4 \Upsilon_{1,i}(\mathbf{q}, \mathbf{p}_i, \dot{\mathbf{p}}_i),$$

$$\tilde{\Gamma}_2(\mathbf{q}, \mathbf{p}, \dot{\mathbf{p}}) = \Gamma_2(\mathbf{q}) + \sum_{i=5}^7 \Upsilon_{2,i}(\mathbf{q}, \mathbf{p}_i, \dot{\mathbf{p}}_i),$$

where

$$\Upsilon_{1,i}(\mathbf{q}, \mathbf{p}_i, \dot{\mathbf{p}}_i) = \exp\left(-\frac{\|\mathbf{p}_i - \mathbf{q}\|^2}{\chi_{1,i}^2}\right) \Lambda_{1,i} \dot{\mathbf{p}}_i(t),$$

$$\Upsilon_{2,i}(\mathbf{q}, \mathbf{p}_i, \dot{\mathbf{p}}_i) = \exp\left(-\frac{\|\mathbf{p}_i - \mathbf{q}\|^2}{\chi_{2,i}^2}\right) \Lambda_{2,i} \dot{\mathbf{p}}_i(t),$$

in which $\Gamma_1(\mathbf{q}) = \Gamma_2(\mathbf{q}) = \begin{bmatrix} 0 & -120 \\ 50 & 0 \end{bmatrix} \cdot \mathbf{q}$, $\chi_{1,1} = 0.2$, $\chi_{1,2} = 0.3$, $\chi_{1,3} = 0.2$, $\chi_{1,4} = 0.4$, $\chi_{2,5} = \chi_{2,6} = \chi_{2,7} = 0.2$, $\Lambda_{1,1} = \text{diag}(150, 100)$, $\Lambda_{1,2} = \text{diag}(200, 50)$, $\Lambda_{1,3} = \text{diag}(150, 50)$, $\Lambda_{1,4} = \text{diag}(200, 100)$, $\Lambda_{2,5} = \Lambda_{2,6} = \Lambda_{2,7} = \text{diag}(150, 70)$; (ii) Regarding the diffusion terms, $D_{11} = D_{22} = 1$, and $D_{12} = D_{21} = 10^{-8}$ as cross-diffusion effects do not arise often in practical biological models [33]; (iii) Regarding the augmented reaction terms, assuming no prey consumed or generated intrinsically, we let $R_1(\mathbf{q}, t) = R_2(\mathbf{q}, t) = 0$, and the influence of the predators be reflected by $f_{n,i}^R = -5 \times 10^3 \cdot \phi_n(\mathbf{q}, t)$, $\delta_{n,i}^R = 10^{-3}$, $\forall i \in \{1, 2, 3, 4, 5, 6, 7\}$, $n \in \{1, 2\}$, and $r_{1,1}^R = 0.02 \times 7$, $r_{1,2}^R = 0.02 \times 6$, $r_{1,3}^R = 0.02 \times 5$, $r_{1,4}^R = 0.02 \times 4$, and $r_{2,5}^R = r_{2,6}^R = r_{2,7}^R = 0.02 \times 7$. Regarding the weighting metrics encoding the relative performance or characteristics among the predators with respect to $\phi_1(\mathbf{q}, t)$, we let $w_{1,1} = 1/7$, $w_{2,1} = 1/6$, $w_{3,1} = 1/5$, and $w_{4,1} = 1/4$. Regarding the ones

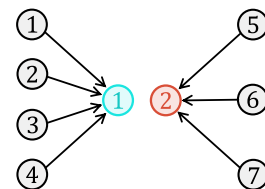


Figure 12. The predator-prey graph for the pursuit-evasion application (the second experiment) in Section VI-B.

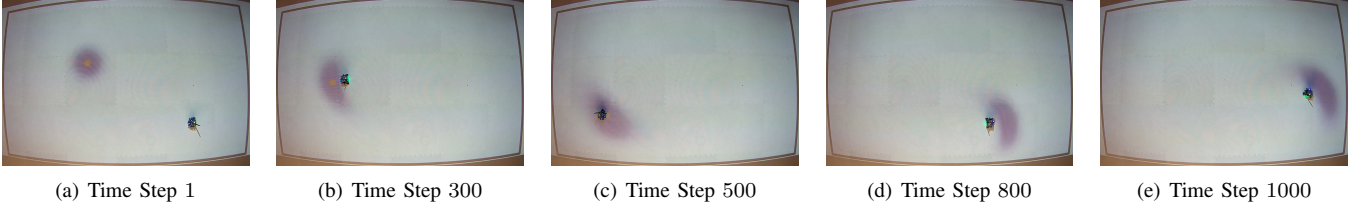


Figure 13. Snapshots of one robot pursuing one prey species represented by the lavender density (without reaction process). The orange dot represents the optimal coverage position of the robot over the prey.

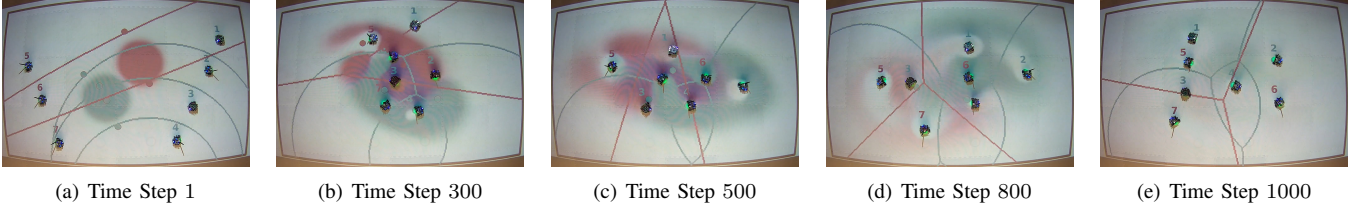


Figure 14. Snapshots of the deployment of seven robots to pursue two prey species. Prey Species 1 represented by the cyan density $\phi_1(\mathbf{q}, t)$ is being pursued by Robots 1, 2, 3, and 4 labeled by cyan numbers, each of which is in charge of a dominant subdomain (Voronoi cell) with a set of cyan curves as its boundaries and a cyan dot as its center of mass with respect to $\phi_1(\mathbf{q}, t)$. Prey Species 2 represented by the red density $\phi_2(\mathbf{q}, t)$ is being pursued by Robots 5, 6, and 7 labeled by red numbers, each of which is in charge of a dominant subdomain (Voronoi cell) with a set of red lines as its boundaries and a red dot as its center of mass with respect to $\phi_2(\mathbf{q}, t)$.

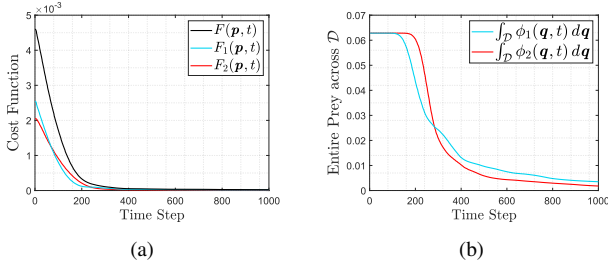


Figure 15. Results of the second experiment of pursuit-evasion in Section VI-B. (a) Evolution of the cost function (15); (b) Evolution of the entire prey ($\phi_1(\mathbf{q}, t)$ and $\phi_2(\mathbf{q}, t)$ separately) across $\mathcal{D}$.

with respect to $\phi_2(\mathbf{q}, t)$, we let $w_{5,2} = w_{6,2} = w_{7,2} = 1/7$. The initial conditions are shown in Fig. 14(a).

The experimental results are shown in Fig. 14 and Fig. 15. According to Fig. 14 and Fig. 15(a), the robots asymptotically achieve a time-varying optimal configuration, and correspondingly the cost function (15) asymptotically decreases, where the cost function is written as $F(\mathbf{p}, t) = F_1(\mathbf{p}, t) + F_2(\mathbf{p}, t)$, where

$$F_1(\mathbf{p}, t) = \sum_{i=1}^4 \int_{\mathcal{V}_{i,1}(\mathbf{p})} w_{i,1}^2 \|\mathbf{p}_i(t) - \mathbf{q}\|^2 \phi_1(\mathbf{q}, t) d\mathbf{q},$$

$$F_2(\mathbf{p}, t) = \sum_{i=5}^7 \int_{\mathcal{V}_{i,2}(\mathbf{p})} w_{i,2}^2 \|\mathbf{p}_i(t) - \mathbf{q}\|^2 \phi_2(\mathbf{q}, t) d\mathbf{q},$$

which verifies Theorem V.1.

According to Fig. 15(b), the total amounts of Prey Species 1 and 2 across $\mathcal{D}$ first stay almost constant and then start to obviously decrease from around time step 200 onward because at the beginning the robots are not close to an optimal coverage over the prey densities so the “consuming” effects have not

been fully realized. From around time step 200 onward, as the robots asymptotically achieve a time-varying optimal coverage over the prey, as also seen in Fig. 14 and Fig. 15(a), both $\int_{\mathcal{D}} \phi_1(\mathbf{q}, t) d\mathbf{q}$ and $\int_{\mathcal{D}} \phi_2(\mathbf{q}, t) d\mathbf{q}$ asymptotically decrease, which verifies Proposition IV.2.

C. Environmental Monitoring

In the application of (32) to pursuit-evasion in Section VI-B, the predators are assumed to have different capabilities of pursuing different prey species, which can be regarded as trivially different task suitability for robots in the sense that each predator’s out-degree in $\mathcal{G}_{\text{prey}}^{\text{pred}}$ is assumed to be 1, resulting in optimal configurations as centroidal Voronoi tessellations, as discussed in Remark V.1.

In this subsection, we consider a more sophisticated (in terms of task suitability) scenario where heterogeneous robots collaborate with each other using their complementary sensing modalities in the sense that some predators’ out-degrees in $\mathcal{G}_{\text{prey}}^{\text{pred}}$ are more than 1, resulting in optimal configurations that, however, are not centroidal Voronoi tessellations in general. We consider this task as a case of multi-agent collaboration, where agents contribute unique or complementary capabilities

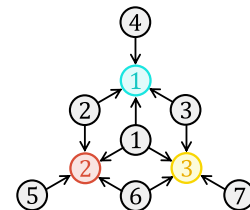


Figure 16. The predator-prey graph for the environmental monitoring application in Section VI-C.

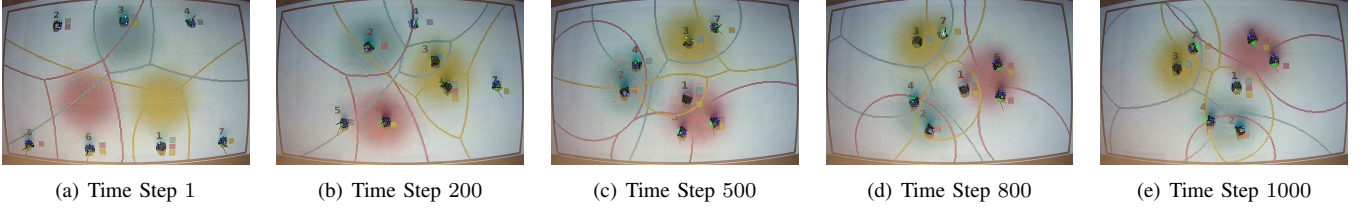


Figure 17. Snapshots of the deployment of seven robots with heterogeneous relative performance or characteristics and complementary sensing modalities to dynamically monitor three qualitatively different time-varying environmental phenomena. $\phi_1(\mathbf{q}, t)$ represented by the cyan density is being monitored by Robots 1, 2, 3, and 4, $\phi_2(\mathbf{q}, t)$ represented by the red density is being monitored by Robots 1, 2, 5, and 6, and $\phi_3(\mathbf{q}, t)$ represented by the yellow density is being monitored by Robots 1, 3, 6, and 7. The robots are labeled by their corresponding numbers nearby, and their sensing modalities are represented by the square(s) nearby them with the same color(s) as the corresponding density (or densities). Each robot is in charge of its dominant subdomain(s) based on its relative performance or characteristic with respect to its target environmental phenomenon (or phenomena), with a set of curves with corresponding color(s) as its (or their) boundaries. The dominant subdomains are quantified by the multiplicatively weighted Voronoi cells. The boundaries of the multiplicatively weighted Voronoi cells of Robots 1, 2, 3, 4 with respect to $\phi_1(\mathbf{q}, t)$ are represented by the cyan curves, the ones of Robots 1, 2, 5, 6 with respect to $\phi_2(\mathbf{q}, t)$ are represented by the red curves, and the ones of Robots 1, 3, 6, 7 with respect to $\phi_3(\mathbf{q}, t)$ are represented by the yellow curves.

to achieve a shared goal that requires the involvement of certain members.

Deploying heterogeneous robots (predators) to monitor (pursue) multiple environmental phenomena (prey) in a complex environment is necessary, where no single sensor type can detect all phenomena. This is because different types of sensors are typically suitable for detecting different types of phenomena. For example, acoustic sensors can be used to monitor fish schools in an environment without good lighting conditions, optical sensors can be used to monitor the migration of zebra herds, temperature sensors be used to monitor forest fire propagation, and chemical sensors be used to monitor the changes of factors vital to the survival of wildlife such as acidity, salinity, humidity, as well as oxygen content of the environment. In addition, it is not efficient, robust, and economical to equip all types of sensors onto every single robot. Thus, the collaboration of heterogeneous robots with complementary sensing modalities becomes important.

Similar to [57], [58], we quantify the environmental phenomena as density distributions to encode the relative importance of the points in the domain of interest, where a point with higher density value corresponds to denser or more important features of phenomena. However, the methods proposed in [57], [58] are designed for monitoring time-invariant environmental phenomena, but many phenomena of interest are time-varying in nature, such as the schooling fish, the propagating forest fire, and the changing acidity, humidity, or oxygen content in the environment. Inspired by [58], the novel coverage control strategy resulting from (32) enables heterogeneous robots to optimally cover various time-varying environmental phenomena. Furthermore, we utilize the multiplicatively weighted Voronoi diagram to encode heterogeneous relative performance or characteristics among robots. In this scenario, a robot may have multiple dominant subdomains (Voronoi cells) with respect to different environmental phenomena (density distributions).

Specifically, we define the predator-prey graph as shown in Fig. 16, which implies that $\mathcal{N} = \{1, 2, 3, 4, 5, 6, 7\}$, $\mathcal{M} = \{1, 2, 3\}$, and the cost function (15) can be written as

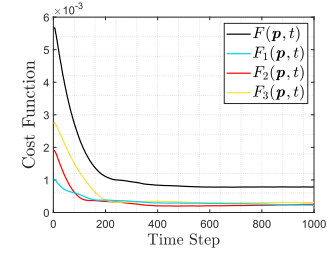


Figure 18. Evolution of the cost function (15) of the environmental monitoring application in Section VI-C.

$F(\mathbf{p}, t) = \sum_{n \in \mathcal{M}} F_n(\mathbf{p}, t)$, where

$$F_n(\mathbf{p}, t) = \sum_{i \in \mathcal{N}_{\mathcal{M}}^n} \int_{\mathcal{V}_{i,n}(\mathbf{p})} w_{i,n}^2 \|\mathbf{p}_i(t) - \mathbf{q}\|^2 \phi_n(\mathbf{q}, t) d\mathbf{q},$$

in which $\mathcal{N}_{\mathcal{M}}^1 = \{1, 2, 3, 4\}$, $\mathcal{N}_{\mathcal{M}}^2 = \{1, 2, 5, 6\}$, and $\mathcal{N}_{\mathcal{M}}^3 = \{1, 3, 6, 7\}$. Regarding the weighting metrics encoding the relative performance or characteristics among the robots with respect to $\phi_1(\mathbf{q}, t)$, we let $w_{1,1} = 1/8$, $w_{2,1} = 1/7$, $w_{3,1} = 1/5$, and $w_{4,1} = 1/6$. Regarding the ones with respect to $\phi_2(\mathbf{q}, t)$, we let $w_{1,2} = 1/8$, $w_{2,2} = 1/5$, $w_{5,2} = 1/6$, and $w_{6,2} = 1/7$. Regarding the ones with respect to $\phi_3(\mathbf{q}, t)$, we let $w_{1,3} = 1/5$, $w_{3,3} = 1/7$, $w_{6,3} = 1/8$, and $w_{7,3} = 1/6$.

In terms of the prey dynamics (9), in order to control variates for the purpose of ablation and comparison experiments with other applications, we omit the intrinsic reaction and diffusion processes of the environmental phenomena. In addition, we simply let the intrinsic drift process of each prey species be a uniform counterclockwise circular motion. Moreover, since typically environmental monitoring robots are supposed to not influence the environmental phenomena, we also omit the augmenting drift and reaction processes. The initial configurations are as shown in Fig. 17(a).

The experimental results are shown in Fig. 17 and Fig. 18, from which one can observe that the robots asymptotically achieve a time-varying optimal configuration, and correspondingly the cost function (15) asymptotically decreases, which verifies Theorem V.1. One can also observe in Fig. 17 the realization of Assumption (c), mentioned at the beginning of

Section V. Note that due to the spatial configuration of the densities defined in this experiment, there are some robots, each of which converges to an optimal position near the center of mass of the dominant subdomain with respect to one of its target densities. For example, as shown in Fig. 17(e), Robot 3 is in charge of both $\phi_1(\mathbf{q}, t)$ and $\phi_3(\mathbf{q}, t)$, with its optimal position as

$$C_3^w(\mathbf{p}, t) = \frac{\sum_{n \in \mathcal{M}_N^3} (w_{3,n}^2 m_{3,n}(\mathbf{p}, t) c_{3,n}(\mathbf{p}, t))}{\sum_{n \in \mathcal{M}_N^3} (w_{3,n}^2 m_{3,n}(\mathbf{p}, t))}$$

in which $\mathcal{M}_N^3 = \{1, 3\}$, and since $m_{3,1}$ is much smaller than $m_{3,3}$ defined by (23) as what can be observed in Fig. 17(e), then $C_3^w(\mathbf{p}, t)$ is very close to $c_{3,3}(\mathbf{p}, t)$ defined by (24). However, in general the optimal position of a robot is not near the center of mass of any of its dominant subdomains, e.g., when the variances of the densities are relatively large.

D. Phoretic Interactions

In Section VI-C, the collaborative pursuit framework (32) is applied to environmental monitoring that requires the collaboration of robots with complementary sensing modalities, where we demonstrate the novel coverage control strategy for different time-varying densities with multiplicatively weighted Voronoi tessellations. In this subsection, this coverage control strategy will also be used in a task that additionally requires the collaboration of robots with different safe operating regions.

Inspired by the phoretic behavior in nature such as remoras riding on larger marine animals [59] and pseudoscorpions on larger insects [60], where the beneficiaries obtain additional mobility by gaining access to expanded operating regions through the facilitators, we divide a team of heterogeneous robots into *facilitators* and *beneficiaries*, and consider the beneficiaries' mobility to be enhanced, reflected by expanded safe operating regions, when in proximity to facilitators.

For example, when ground robots need to reach an island for certain tasks but cannot operate safely in the water, aquatic robots can approach the ground robots. Then, they can "step on" the facilitation regions provided by the aquatic robots and then be transported to the island to perform their main tasks. Another example is on minimally invasive robotic systems. When a surgical robotic arm equipped with an end-effector performs tasks such as tissue manipulation, cutting, or suturing, a sensing arm equipped with perception modules should be close to the end-effector to provide real-time perception feedback of the ambient environment, thereby indirectly expanding the safe operating region of the main surgical robotic arm., e.g., [61]. The examples suggest that the facilitation regions can be physical, as in ground-aquatic transport, or functional, as in surgical robotic collaboration.

Specifically, the safe operating region of a robot is encoded by the corresponding CBFs, such as each robot being constrained in $\mathcal{D}$ through the constraints induced by the set of CBFs (33), (34), (35), and (36). We denote $\mathcal{N}_f \subset \mathcal{N}$ as the set of indices of the facilitators and $\mathcal{N}_b \subset \mathcal{N}$ as the set of indices of the beneficiaries such that $\mathcal{N}_f \cap \mathcal{N}_b = \emptyset$ and $\mathcal{N}_f \cup \mathcal{N}_b = \mathcal{N}$. In addition to the five CBFs (33), (34), (35), (36), and (37) defined for every robot for all time, we denote the CBFs

encoding the (additionally considered) safe operating regions for facilitators as h_i^f , $\forall i \in \mathcal{N}_f$, and the ones for beneficiaries as h_i^b , $\forall i \in \mathcal{N}_b$. Furthermore, we define the facilitation region provided by Robot $j \in \mathcal{N}_f$ (facilitator) for Robot $i \in \mathcal{N}_b$ (beneficiary) as the 0-superlevel set of $h_{i,j}^C$. The CBF $h_{i,j}^C$ can expand the safe operating region of the beneficiary through disjunction of CBFs. According to [62], the conjunction and disjunction of the CBFs h_1 and h_2 are defined as $h_1 \wedge h_2 := \min\{h_1, h_2\}$ and $h_1 \vee h_2 := \max\{h_1, h_2\}$, respectively. As a result, the safe operating region of a beneficiary can be encoded by the CBF $h_{i,6} = h_i^b \vee h_{i,j}^C$, $\forall j \in \mathcal{N}_f^i$, $\forall i \in \mathcal{N}_b$, in which $\mathcal{N}_f^i \subseteq \mathcal{N}_f$ denotes the set of indices of facilitators potentially to facilitate Robot $i \in \mathcal{N}_b$.

We divide $\mathcal{D}$ into three rectangular subregions, the one on the left, $\mathcal{S}_l$, the one in the middle, $\mathcal{S}_m$, and the one on the right, $\mathcal{S}_r$, as illustrated in Fig. 20, where

$$\mathcal{S}_l = \{(x, y) \in \mathbb{R}^2 \mid -1.6 \leq x \leq -0.7 \text{ and } -1 \leq y \leq 1\},$$

$$\mathcal{S}_m = \{(x, y) \in \mathbb{R}^2 \mid -0.7 \leq x \leq 0.7 \text{ and } -1 \leq y \leq 1\},$$

$$\mathcal{S}_r = \{(x, y) \in \mathbb{R}^2 \mid 0.7 \leq x \leq 1.6 \text{ and } -1 \leq y \leq 1\}.$$

Accordingly, we define the CBF that encodes the safe operating region of Robot $i \in \mathcal{N}_f$ (facilitator) as $h_{i,6}(\mathbf{p}_i) = h_i^{f,l}(\mathbf{p}_i) \wedge h_i^{f,r}(\mathbf{p}_i)$, $\forall i \in \mathcal{N}_f$, in which $h_i^{f,l}(\mathbf{p}_i) = [1, 0] \cdot \mathbf{p}_i + 0.7$ and $h_i^{f,r}(\mathbf{p}_i) = 0.7 - [1, 0] \cdot \mathbf{p}_i$. Likewise, using the methods in [62], [63], we set the CBF for Robot $i \in \mathcal{N}_b$ (beneficiary), adapted to the particulars of our problem, as

$$h_{i,6}(\mathbf{p}_i, t) = h_i^b(\mathbf{p}_i) \vee h_{i,j}^C(\mathbf{p}_i, t), \quad \forall i \in \mathcal{N}_b, \quad (39)$$

where $h_i^b(\mathbf{p}_i) = h_i^{b,l}(\mathbf{p}_i) \vee h_i^{b,r}(\mathbf{p}_i)$, in which $h_i^{b,l}(\mathbf{p}_i) = -0.7 - [1, 0] \cdot \mathbf{p}_i$ and $h_i^{b,r}(\mathbf{p}_i) = [1, 0] \cdot \mathbf{p}_i - 0.7$, and

$$h_{i,j}^C(\mathbf{p}_i, t) = C_{i,j}^2 - \|\mathbf{p}_i - \mathbf{p}_j(t)\|^2, \quad \forall j \in \mathcal{N}_f^i,$$

in which $C_{i,j} \in \mathbb{R}_{>0}$ denotes the collaboration distance between Robot $i \in \mathcal{N}_b$ and Robot $j \in \mathcal{N}_f$, to encode disjunction that Robot $i \in \mathcal{N}_b$ should stay in the safe sets quantified by $h_i^b(\mathbf{p}_i)$ or $h_{i,j}^C(\mathbf{p}_i, t)$. Note that the Clarke generalized gradient should be applied when utilizing the CBF composed by Boolean operations, and we refer to [62], [63] for detailed methods to analyze the Boolean compositions of CBFs.

To control variates for ablation and comparison experiments, we perform an experiment on the Robotarium with each facilitator able to pursue the time-varying density induced by every beneficiary, i.e., each facilitator has the potential to facilitate every beneficiary. Hence, we let $\mathcal{N}_f^i = \mathcal{N}_f$, $\forall i \in \mathcal{N}_b$, and correspondingly, we define the predator-prey graph as

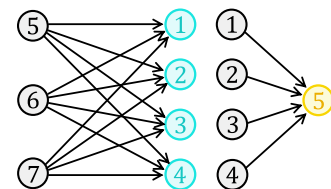


Figure 19. The predator-prey graph for the physical experiment of the application of phoretic interactions in Section VI-D.

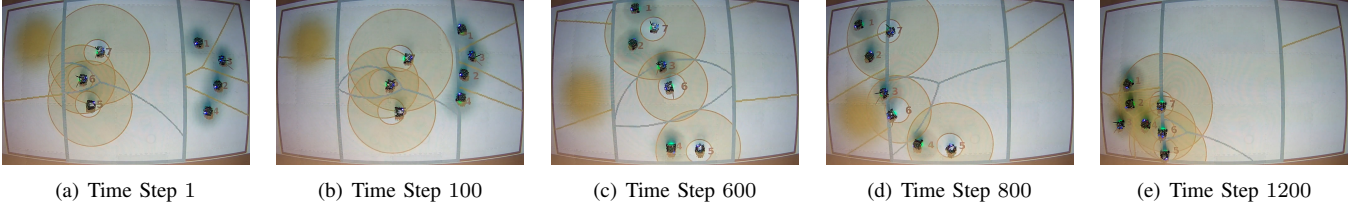


Figure 20. Snapshots of the physical experiment of the application of phoretic interactions. Robots 1, 2, 3, and 4 are the beneficiaries and Robots 5, 6, and 7 are the facilitators. (Each robot is labeled with its corresponding number nearby.) The beneficiaries can safely operate in $\mathcal{S}_r$, $\mathcal{S}_l$, or the facilitation regions provided by the facilitators. Each translucent yellow annulus represents the facilitation region provided by the corresponding facilitator excluding the collision avoidance region. The facilitators can safely operate in $\mathcal{S}_m$. Each facilitator is covering the cyan densities $\phi_1(\mathbf{q}, t)$, $\phi_2(\mathbf{q}, t)$, $\phi_3(\mathbf{q}, t)$, and $\phi_4(\mathbf{q}, t)$ induced by the states of the beneficiaries, using the coverage control strategy demonstrated in Section VI-C. The boundaries of the Voronoi cells of the facilitators are represented by the cyan curves. The weight of each Voronoi cell with respect to each cyan density is related to its corresponding collaboration distance $C_{i,j}$ to encode that the larger the facilitation region, the larger the dominant subdomain of the corresponding facilitator. The beneficiaries are covering the yellow density $\phi_5(\mathbf{q}, t)$, and the corresponding boundaries of the Voronoi cells are shown by the yellow lines.

shown in Fig. 19, and let $\mathcal{N}_b = \{1, 2, 3, 4\}$, $\mathcal{N}_f = \{5, 6, 7\}$, $\mathcal{M} = \{1, 2, 3, 4, 5\}$. Note that the assumption that each facilitator has the potential to facilitate every beneficiary does not mean that it is hard-coded which facilitators should be assigned to collaborate with exactly which beneficiaries in advance. The collaboration process occurs autonomously using the proposed collaborative pursuit framework (32), so that the robots can adapt to changing conditions.

According to Fig. 19, the cost function (15) can be written as $F(\mathbf{p}, t) = \sum_{n \in \mathcal{M}} F_n(\mathbf{p}, t)$, where

$$F_n(\mathbf{p}, t) = \sum_{i \in \mathcal{N}_{\mathcal{M}}^n} \int_{\mathcal{V}_{i,n}(\mathbf{p})} w_{i,n}^2 \|\mathbf{p}_i(t) - \mathbf{q}\|^2 \phi_n(\mathbf{q}, t) d\mathbf{q}, \quad (40)$$

in which $\mathcal{N}_{\mathcal{M}}^n = \{5, 6, 7\}$, $\forall n \in \{1, 2, 3, 4\}$, and $\mathcal{N}_{\mathcal{M}}^5 = \{1, 2, 3, 4\}$. Regarding the weighting metrics encoding the relative performance or characteristics among the robots with respect to $\phi_n(\mathbf{q}, t)$, we let $w_{5,n} = 1/5$, $w_{6,n} = 1/4$, and $w_{7,n} = 1/6$, $\forall n \in \{1, 2, 3, 4\}$, to specifically encode the size of the facilitation region such that the larger the facilitation region a facilitator provides for a beneficiary, the larger the dominant subdomain of this facilitator with respect to the density induced by this beneficiary among all the facilitators that pursues this density. Thus, we let $C_{i,5} = 0.5$, $C_{i,6} = 0.4$, $C_{i,7} = 0.6$, $\forall i \in \mathcal{N}_b$. In addition, we assume Robots 1, 2, 3, and 4 are homogeneous with respect to $\phi_5(\mathbf{q}, t)$ and thus let $w_{1,5} = w_{2,5} = w_{3,5} = w_{4,5} = 1/5$. For the intrinsic drift process in (9), we let $\Gamma_1(t) = \dot{\mathbf{p}}_1(t)$, $\Gamma_2(t) = \dot{\mathbf{p}}_2(t)$, $\Gamma_3(t) = \dot{\mathbf{p}}_3(t)$, $\Gamma_4(t) = \dot{\mathbf{p}}_4(t)$, $\Gamma_5(t) = [0, \cos(\omega t)/2]^\top$. To control variates, we omit all the influences from the predators, as well as the intrinsic diffusion and reaction processes. The initial conditions are as shown in Fig. 20(a).

The results of the physical experiment are shown in Fig. 20 and Fig. 21. At the beginning, the beneficiaries are initialized in $\mathcal{S}_r$ and try to optimally cover their target prey density $\phi_5(\mathbf{q}, t)$, the yellow density mainly distributed in $\mathcal{S}_l$. However, since the beneficiaries are prevented by the CBF-induced constraints from going across $\mathcal{S}_m$, they are stuck at the right-hand-side boundary of $\mathcal{S}_m$ right before around time step 100 as shown in Fig. 20(b), which is why the overall deceleration of cost $F_5(\mathbf{p}, t)$ decreases around time step 100 as shown in Fig. 21(b). Driven by the coverage control strategy that is

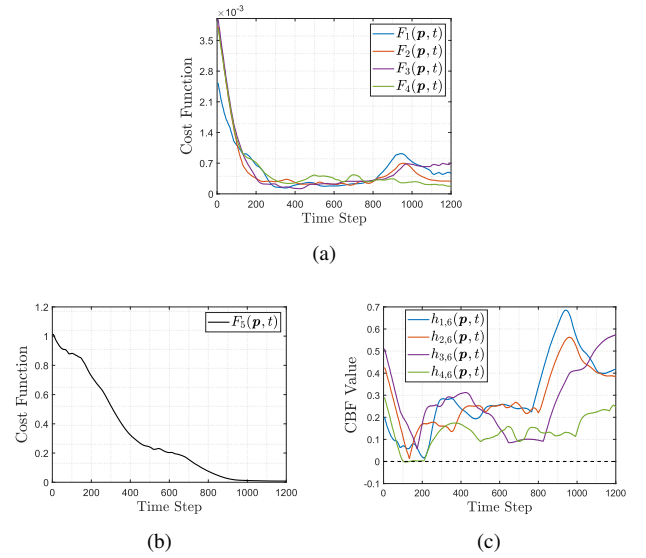


Figure 21. Results of the physical experiment of phoretic interactions in Section VI-D. (a) Evolution of the cost components $F_1(\mathbf{p}, t)$, $F_2(\mathbf{p}, t)$, $F_3(\mathbf{p}, t)$, and $F_4(\mathbf{p}, t)$ given by (40); (b) Evolution of the cost component $F_5(\mathbf{p}, t)$ per (40); (c) Evolution of the values of the CBFs given by (39).

also implemented in Section VI-C, the facilitators approach the beneficiaries. Then, the safe operating regions of the beneficiaries are expanded through the facilitation regions provided by the facilitators so that they can traverse $\mathcal{S}_m$ to reach $\mathcal{S}_l$, and then optimally cover their target prey density, resulting in the ongoing decrease of $F_5(\mathbf{p}, t)$. From around time step 450 to 650, the overall deceleration of the cost $F_5(\mathbf{p}, t)$ decreases due to the constraints of the facilitation regions imposed on the beneficiaries. Additionally, Fig. 21(a) shows that the cost components $F_1(\mathbf{p}, t)$, $F_2(\mathbf{p}, t)$, $F_3(\mathbf{p}, t)$, and $F_4(\mathbf{p}, t)$ tend to increase after around time step 800 as the facilitators are stuck at the left-hand-side boundary of $\mathcal{S}_m$ by their CBF-induced constraints. However, since we focus on the transport of beneficiaries to $\mathcal{S}_l$ where their target prey densities are mainly distributed, it is acceptable that the cost components of the facilitators even increase after the successful completion of the mainly concerned task. Specifically, we consider the success of the primary task as the situation in which all the beneficiaries are transported to $\mathcal{S}_l$. Furthermore, the value of

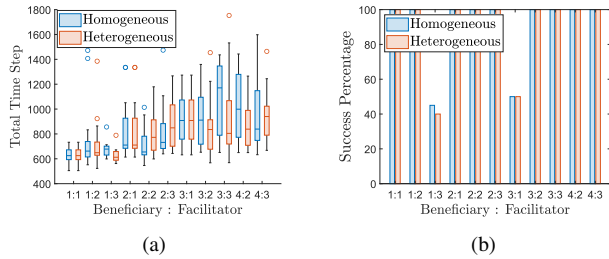


Figure 22. Statistical results of the simulation experiments of the phoretic interactions in Section VI-D. (a) The box plot of the final time step (based on successful trials) when all the beneficiaries are transported to S_1 where their target prey density is mainly distributed; (b) The corresponding bar plot of the success rate for each ratio of facilitators to beneficiaries with homogeneous facilitators and heterogeneous facilitators respectively.

the CBF (39) for each beneficiary is plotted in Fig. 21(c), from which one can observe that each CBF value stays nonnegative, which means that all beneficiaries stay safe. In addition, Fig. 20 shows that all facilitators also remain safe.

In addition to the physical experiment, to further evaluate the performance of the phoresis strategy resulting from the proposed collaborative pursuit framework (32), we conduct 380 trials of simulation experiments, of which 40 trials for each ratio of facilitators to beneficiaries with homogeneous facilitators and heterogeneous facilitators, i.e., 20 trials for homogeneous facilitators and 20 trials for heterogeneous facilitators. If the number of facilitator is 1, we perform 20 trials and plot the identical results still for homogeneous and heterogeneous categories for ease of comparisons.

The initial positions of the beneficiaries are randomly generated within $[x_{\min}, x_{\max}] = [0.9, 1.4]$ and $[y_{\min}, y_{\max}] = [-0.8, 0.8]$, and the ones of the facilitators are randomly generated within $[x_{\min}, x_{\max}] = [-0.5, 0.5]$ and $[y_{\min}, y_{\max}] = [-0.8, 0.8]$. Moreover, we let $w_{5,n} = w_{6,n} = w_{7,n} = 1/5$, $\forall n \in \{1, 2, 3, 4\}$ and $C_{i,5} = C_{i,6} = C_{i,7} = 0.5$, $\forall i \in \mathcal{N}_b$ for the case with homogeneous facilitators in simulations. All other settings are the same as the physical experiment conducted on the Robotarium for fair comparisons.

The statistical results of the simulation experiments are shown in Fig. 22. It can be observed that with the increase in the total number of robots, the final time step when all the beneficiaries are transported to S_1 and the spread of the boxes exhibit an overall increasing tendency, which is likely due to the CBF-induced constraints affecting the overall process and introducing more variability in performance. In the trials with a 1 : 3 ratio of beneficiary to facilitators, the success rates are 45% and 40% for cases with homogeneous and heterogeneous facilitators, respectively. Here, the success rate is defined as the percentage of trials in which the beneficiaries are all transported to S_1 out of the total number of trials. The lower success rate (compared with the trials with 100% success rate) is due to the collision avoidance constraints that can lead to deadlocks, which is still an open problem in the field of CBF. This is reasonable because a certain size of experimental platform accommodates a specific number of robots, especially when considering the collision avoidance requirement. In the trials with a 3 : 1 ratio of beneficiaries

to facilitator, the success rate is 50%, which is because the facilitator may leave one or more beneficiaries in S_r . One can postulate that the performance would be worse if the ratio of beneficiaries to facilitators is larger, which is also reasonable because a certain number of facilitators support a specific number of beneficiaries. This situation can be alleviated by increasing the number of facilitators, as observed from the trials with 3 : 2 and 3 : 3 ratios of beneficiaries to facilitator.

In summary, using the phoresis strategy resulting from the collaborative pursuit framework (32), the collaborations happen autonomously without commanding each robot exactly what to perform before deployment. For example, which facilitator should collaborate with which beneficiary in practice, the sequence of collaborations, whether multiple facilitators collaborate with one beneficiary interchangeably or simultaneously, and how many beneficiaries that one facilitator should collaborate with, can all be handled autonomously by the obtained phoresis strategy.

VII. CONCLUSIONS

In this paper, we developed a general multi-robot collaborative pursuit framework that is adaptive to both dynamic environments and diverse tasks, with adaptivity interpreted as runtime responsiveness to environmental changes and structural flexibility across various tasks. Its effectiveness is demonstrated through four experimental applications: forest firefighting, pursuit-evasion, environmental monitoring, and phoretic interactions. This adaptivity enables a wide range of applications, as evidenced by the examples in this paper, among other potential uses. To name a few, positioning environmental cleaning robots for pollutant removal, deploying multi-robot teams for monitoring eroded beaches, coral reefs, and underwater regions, gathering information with mobile robots for environmental exploration, and portraying dynamic pattern formations in drone art shows.

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